

Immersive Studio Pro

User manual

Video editing for fulldome domes, flat screens and 360° rooms.

Contents

1	Immersive Studio Pro	6
1.1	Formats	6
1.2	Conventions in this manual	6
2	Installing and starting	7
2.1	Requirements	7
2.2	Installation	7
2.3	The start screen	7
2.4	Demo projects	8
2.5	Settings	8
3	How the program is organised	9
3.1	The composite master	9
3.2	Dome coverage	9
3.3	Full dome source and placed clips	10
3.4	Sequences are media	11
3.5	Clips	11
4	A first project	12
4.1	Create the sequence	12
4.2	Import footage	12
4.3	Place a clip on the dome	12
4.4	Add movement	12
4.5	Fill the dome with a composition	13
4.6	Export a master	13
5	The main window	14
5.1	Panels	14
5.2	The command palette	15
5.3	Preferences	16
5.4	Output to a projector	16
6	The media panel	17
6.1	Search	18
6.2	Folders	18
6.3	Badges	19
6.4	Selecting	19

7	The viewer	20
7.1	2D and 3D	20
7.2	Working in the viewer	20
8	The timeline	21
8.1	Track headers	21
8.2	Tools	22
8.3	The transport	22
8.4	Zoom and navigation	23
8.5	Markers	23
9	The inspector	24
9.1	Parameter rows	25
9.2	Header	25
10	Importing and managing media	26
10.1	Import routes	26
10.2	Proxies	26
10.3	The source monitor	26
10.4	Relinking and replacing	27
10.5	Text and shapes	28
10.6	Saving	28
11	Editing in the timeline	29
11.1	Moving clips	29
11.2	Trimming	29
11.3	Fades and crossfades	29
11.4	Cutting and deleting	29
11.5	Copy and paste	30
11.6	Selection and snapping	30
11.7	Speed and looping	31
11.8	The clip menu	32
12	Placing clips in the dome	33
12.1	Transform	33
12.2	Shape on a curved surface	33
12.3	Source	34
12.4	Adjustment layers	35
13	2D sequences	36
13.1	Differences	36
13.2	2D inside dome work	36

14	The 360° room	37
14.1	Geometry	37
14.2	The strip	38
14.3	The 3D view	38
14.4	Exporting a room	39
15	Sequences and nests	40
15.1	Tabs	40
15.2	Creating a sequence	41
15.3	Nesting	41
15.4	Composition proxies	42
15.5	Render in place	42
16	Compose	43
16.1	Building a composition	43
16.2	Types	44
16.3	Tunnel	45
16.4	Weave	47
16.5	Editing a composition	48
17	Effects and masks	49
17.1	The Clip section	49
17.2	Masks	50
17.3	The effects chain	50
18	Colour	51
18.1	Sliders	51
18.2	Lift, Gamma, Gain	52
18.3	Curves	52
18.4	LUTs	52
19	Motion, keyframes and automation	53
19.1	Motion	53
19.2	Keyframes	54
19.3	The automation lane	55
19.4	Mixing motion and automation	55
19.5	Modulation	55
20	Reactive FX	56
20.1	The audio engine	56
20.2	An effect in the chain	57

21	Audio	58
21.1	Tracks and clips	58
21.2	The audio inspector	58
21.3	Fades	58
21.4	Linked audio and video	58
21.5	Audio in exports	58
21.6	Beat markers	58
22	Exporting	59
22.1	Codecs	60
22.2	Quality and bit depth	60
22.3	Presets	60
22.4	The queue	60
23	Corner data	61
23.1	Positions	61
23.2	Viewing and recording	61
23.3	Details	62
24	Live output	63
24.1	NDI input	63
25	Menus	64
25.1	File	64
25.2	Edit	65
25.3	Project	65
25.4	Window	65
26	Keyboard shortcuts	66
27	Compose parameters	68
27.1	Dome types	68
27.2	Flat types	68
27.3	Common parameters	68
28	Effects	70
29	Performance and problem solving	72
29.1	Smooth playback	72
29.2	Situations	72
29.3	Recovering work	73
29.4	Reporting a problem	73

Immersive Studio Pro

Immersive Studio Pro edits video for surfaces that surround an audience: fulldome domes, flat screens, and rooms whose walls and floor are all projection surfaces.

The program runs on the desktop, on Windows and macOS. All rendering happens on the GPU: clips are warped onto the dome in real time, effects run as shaders, and every visible clip is composited into one master image each frame. The viewer displays that master, the live outputs broadcast it, and the exporter records it.

1.1 Formats

A sequence works in one of three modes, chosen when it is created.

Dome

A circular fisheye image covering a hemisphere. Clips are placed by direction — azimuth and elevation — and warped onto the dome surface. The master is square and the projected image is the circle inscribed in it.

2D

A rectangular canvas. Clips are placed by horizontal position, vertical position and scale.

360 Room

A set of walls with an optional floor. The walls are unrolled into one wide strip that is edited as a rectangular canvas, and a 3D view shows the result from inside the room.

Formats mix inside one project: a dome project can contain 2D sequences, and a 2D sequence can be placed inside a dome sequence as a clip.

1.2 Conventions in this manual

Interface labels appear as they read on screen: **Fulldome src**. Keys appear as **Ctrl**+**E**; on macOS, **Ctrl** means **⌘**. Orange numbers on a screenshot correspond to the numbered list below it.

The interface is in English, with a Spanish translation available in the preferences.

Installing and starting

2.1 Requirements

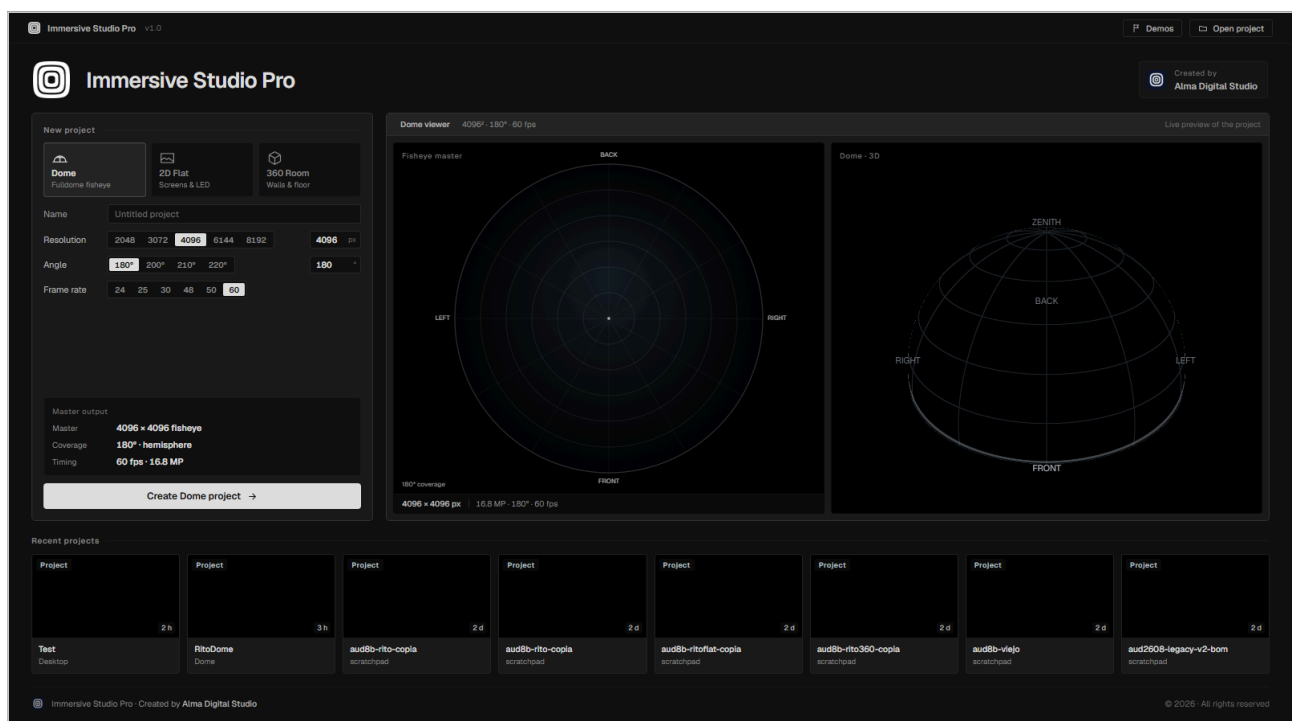
Windows	Windows 10 or 11, 64-bit. On hybrid-graphics laptops the program requests the discrete GPU automatically.
macOS	Apple Silicon.
Memory	16 GB for dome work at 4096×4096 . Decoding several layers of high-bitrate footage at once is the demanding part.
Disk	Local storage for media. A 4096×4096 export at 200 Mb/s occupies about 300 MB per minute.

2.2 Installation

On Windows, run **Immersive Studio Pro Setup 1.0.0.exe**. It installs for the current user, so it requires no administrator rights, and it registers the **.isp** extension so that double-clicking a project opens it.

2.3 The start screen

The program opens on the start screen, where a project is created. The format is chosen first, then its parameters, and a preview shows the result before the project exists.



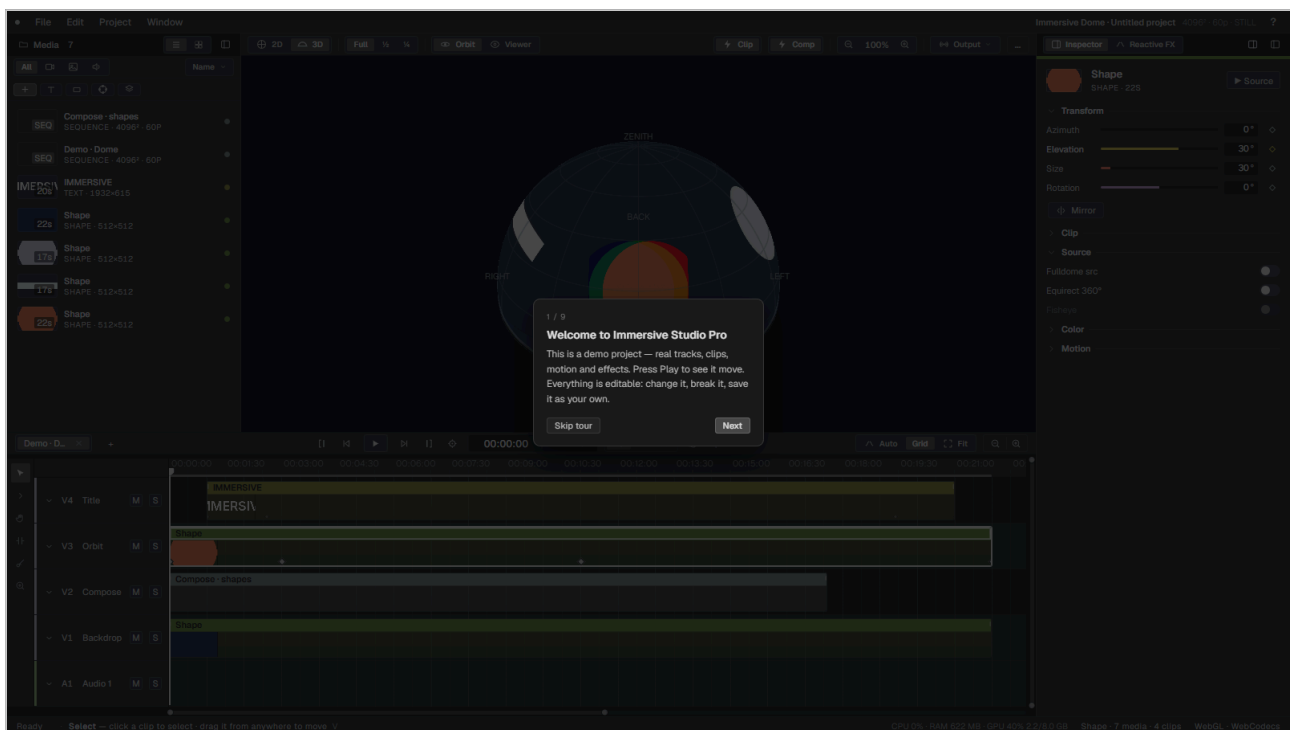
The start screen.

1. Choose **Dome**, **2D** or **360 Room**.
2. Set the parameters: resolution, frame rate, dome coverage, or wall geometry for a room. The preview follows each change.
3. Create the project. The editor opens with an empty timeline.

The start screen is reachable at any time from **File** → **New project...** (**Ctrl** + **N**). An open project remains open behind it, and **Back to project** returns to it unchanged.

2.4 Demo projects

Demos in the top bar builds a complete project in the chosen format — several tracks, clips, a composition, motion, an effect and automation — and runs a guided tour over it. Everything in it is editable and can be saved as a new project.



The guided tour highlights one region at a time. **→** and **←** move through it, **Esc** ends it. It can be restarted from **Window** → **Guided tour**.

2.5 Settings

Application settings, including the interface language, are in **Project** → **Preferences...** Settings that belong to the sequence — resolution, frame rate, dome coverage — are in **Project** → **Sequence settings...**

How the program is organised

Five ideas describe most of the program's behaviour.

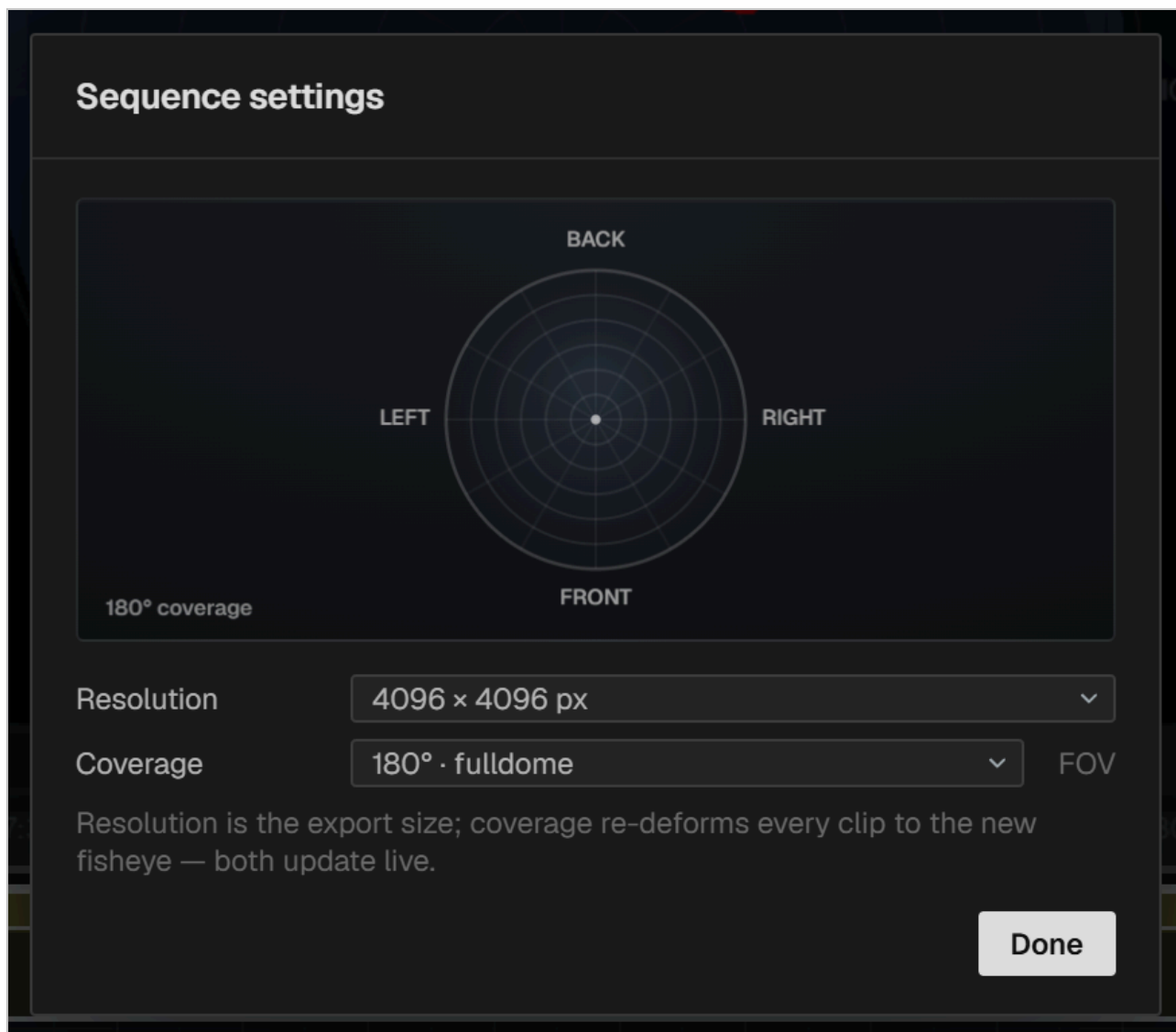
3.1 The composite master

Every visible clip is drawn into a single master image each frame. The viewer shows that master, NDI and Spout broadcast it, and the exporter records it. The image on screen while editing is the image that is delivered.

In a dome sequence the master is square and the picture is the circle inscribed in it. The four corners fall outside the projected area: they are recorded as black, or as transparency when the export carries alpha.

3.2 Dome coverage

A fulldome master is an azimuthal-equidistant fisheye. Its **coverage** is the field of view it represents: 180° for a hemisphere, or 200°, 210° and 220° for domes that continue below the horizon. The setting is in **Project → Sequence settings...** and applies everywhere at once — the warp, the guides, the 3D view and the export.



Sequence settings. Coverage updates live.

Coverage describes the venue. A master made at 180° and projected in a 210° dome is wrong at the edges, and repositioning clips does not correct it. The value comes from the dome's specification.

3.3 Fulldome source and placed clips

A clip in a dome sequence is treated in one of two ways, set by **Fulldome src** in the inspector.

Switched off — a placed clip

The footage is a rectangular picture. It occupies a direction and a size on the dome, and the GPU bends it to sit on the curved surface. This applies to material that was not shot for a dome.

Switched on — a fulldome source

The footage is already a dome master: a circular fisheye filling the frame. It is drawn into the master one to one, since it already covers the whole sky. Generated compositions are fulldome sources.

Equirect 360° handles equirectangular 360° footage and reprojects it into the dome. **Fisheye**, with its **Amount**, bends a fulldome source further; it becomes available once **Fulldome src** is on.

3.4 Sequences are media

A sequence is an item in the media panel, and an item in the media panel can be placed on a track. One mechanism therefore provides sequence tabs, nesting and generated compositions. Chapter 15 describes it.

3.5 Clips

Video, stills, audio, text, shapes, generated compositions, nested sequences and adjustment layers are all clips on tracks. They trim alike, carry the same inspector sections where those apply, and animate alike.

A first project

This walkthrough creates a dome sequence, imports footage, places it, animates it, adds a generated composition and exports a master.

4.1 Create the sequence

1. On the start screen choose **Dome**.
2. Set the resolution to 4096×4096 and the frame rate to the value the venue uses.
3. Set coverage to the dome's field of view.
4. Create the project.

4.2 Import footage

Drag files onto the media panel, or press **Ctrl+I**. Video, stills, audio and image sequences are accepted. For heavy footage, right-click the item and choose **Generate proxy**; chapter 10 covers proxies.

4.3 Place a clip on the dome

1. Drag a clip from the media panel onto a video track.
2. Select it and open **Transform** in the inspector.
3. Drag **Azimuth** to swing it around the horizon, **Elevation** to raise it towards the zenith, and **Size** to enlarge it.

Dragging the clip in the viewer moves the same two values.

Elevation accepts negative values, which place a clip below the horizon. In a dome with coverage above 180° that is projected area.

4.4 Add movement

Open **Motion** in the inspector and click a preset: **Orbit** travels around the horizon, **Float** drifts slowly. A preset adds ordinary modifiers to the list, which can then be retuned, switched off or deleted one at a time.

For a movement with an exact start and end, use keyframes: place the playhead, set the value, and a keyframe is recorded. Chapter 19 covers both routes.

4.5 Fill the dome with a composition

1. Select two or three items in the media panel.
2. Click **Compose** in the create row.
3. Choose a type — **Ring** is the simplest — and set the count.
4. Confirm. The composition arrives on the timeline as one clip.

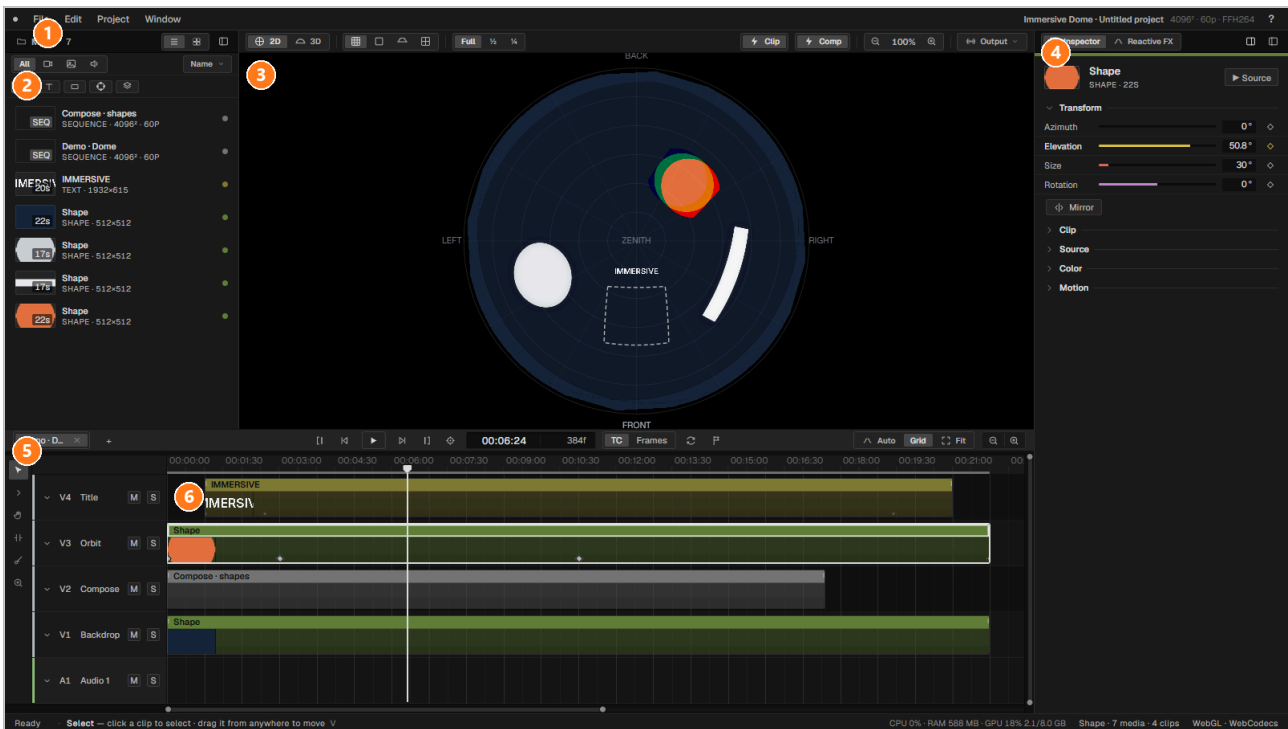
The composition is a nested sequence containing real clips. Double-clicking it opens its contents for editing, and its generating parameters remain in the inspector.

4.6 Export a master

1. Open **File** → **Export...** (**Ctrl** + **Shift** + **E**).
2. Choose the destination folder with **Choose....**
3. Select **MP4 H.264 · GPU, up to 4096**.
4. Set **Quality** to **Maximum** and check the bitrate.
5. Press **Export**.

Projects are saved with **Ctrl** + **S** as .isp files.

The main window



The main window, in dome mode.

- 1 **Menu bar** — File, Edit, Project and Window. Chapter 25 lists every entry. At the right of the same bar, the format chip shows the active sequence's format, resolution, frame rate and export codec, and opens the sequence settings.
- 2 **Media panel** — everything the project contains. Chapter 6.
- 3 **Viewer** — the master image, in 2D or 3D. Chapter 7.
- 4 **Inspector** — the selected clip's parameters. Chapter 9.
- 5 **Transport** — playback controls, timecode, sequence tabs and the editing well. Chapter 8.
- 6 **Timeline** — tracks, clips, ruler, markers and automation lanes, with the tool rail down its left edge. Chapter 8.

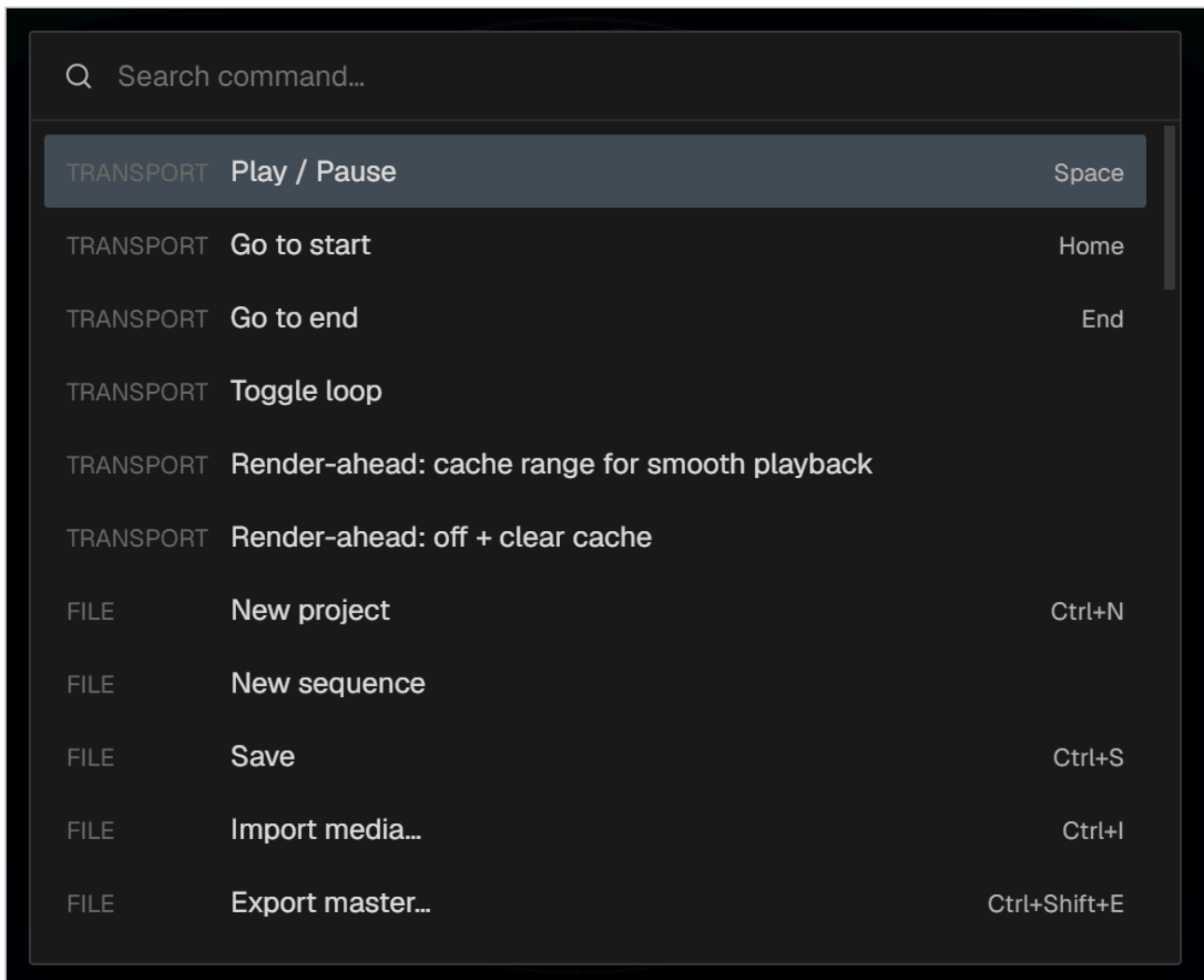
The status bar along the foot of the window describes what the active tool does, and shows CPU, memory and GPU load with the project's media and clip counts.

5.1 Panels

Dragging the gutters between panels resizes them. The media and inspector panels collapse from **Window**, or from the two buttons at the top right of the inspector. The timeline's height is dragged from its top edge.

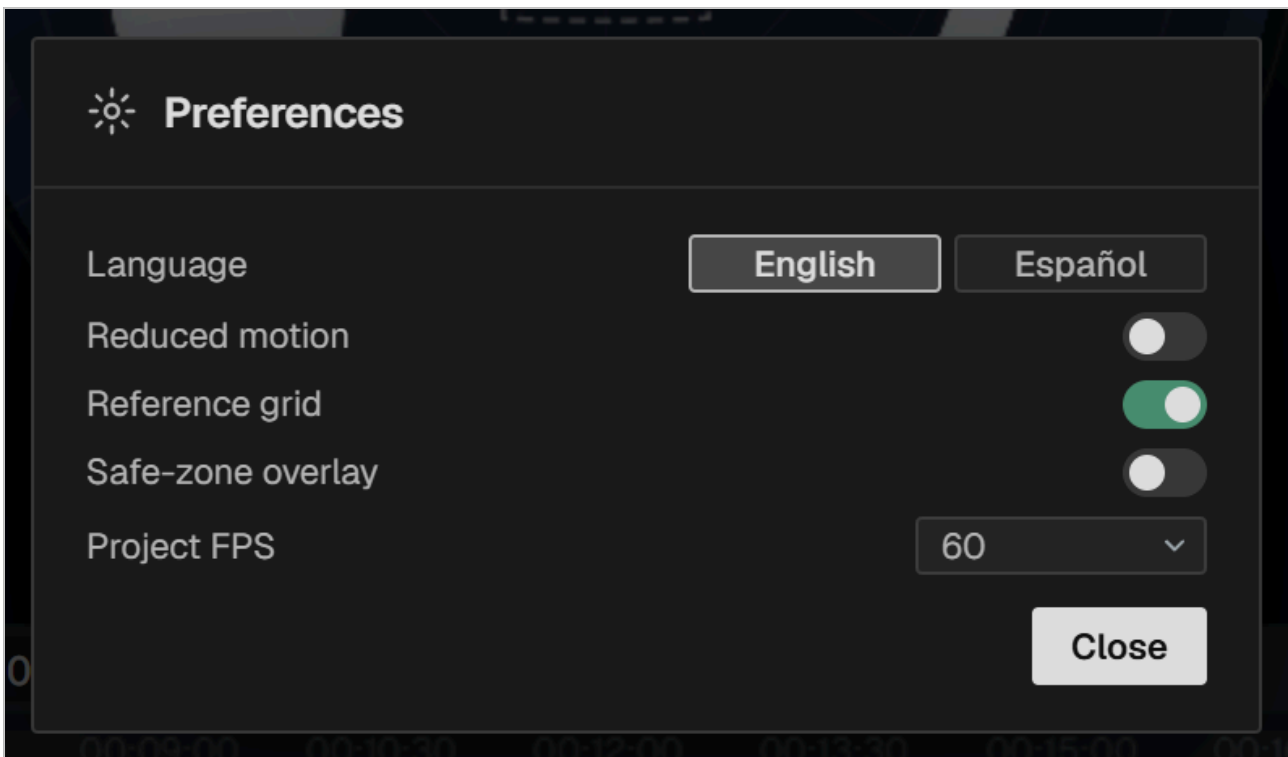
5.2 The command palette

Ctrl + **K**, **F1** or **?** opens a searchable list of every command with its shortcut.



The command palette. Typing filters, the arrow keys move, **Enter** runs.

5.3 Preferences



Preferences, including the interface language.

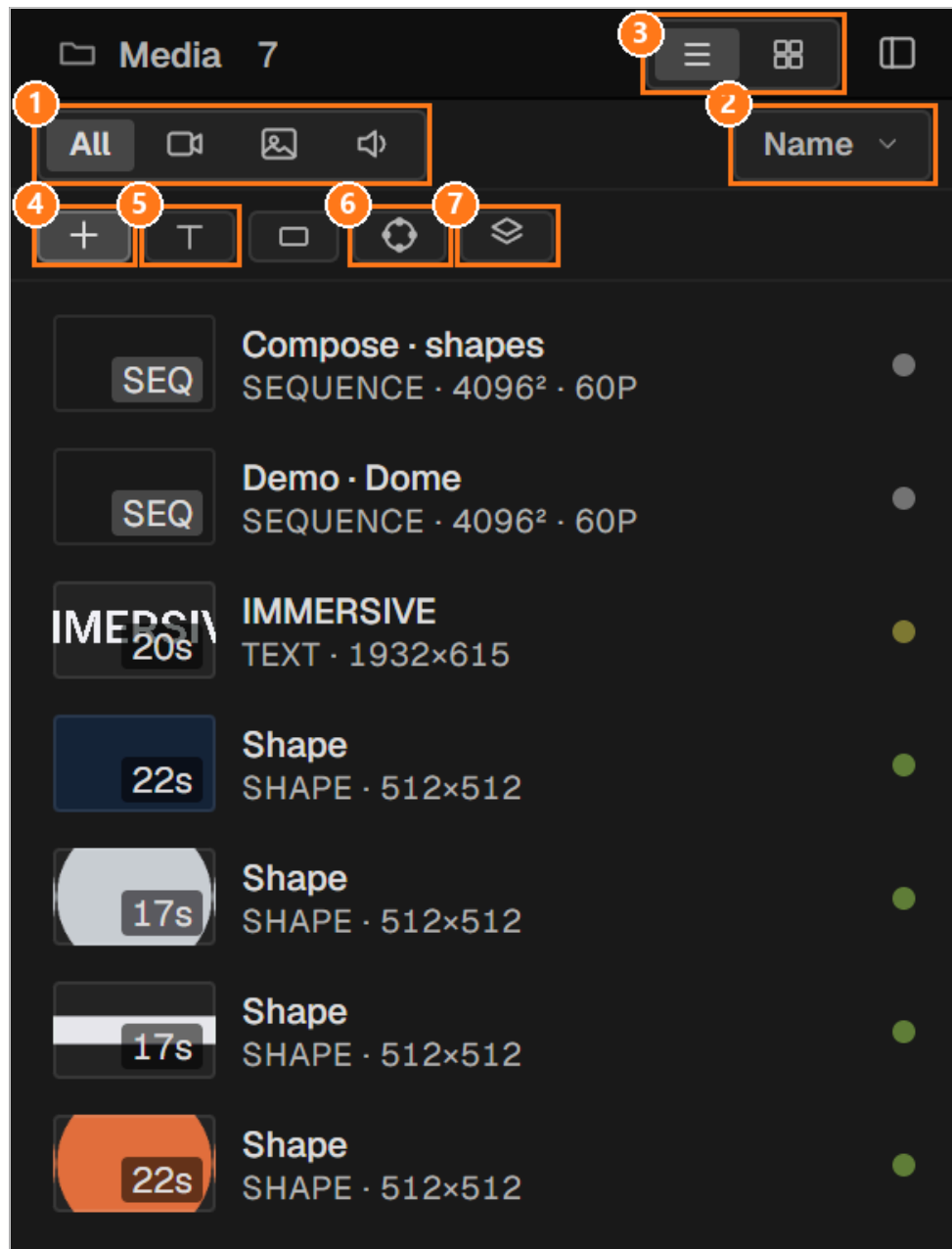
5.4 Output to a projector

Two arrangements send the picture to a display without the interface around it, both under **Window** and in the **Output** menu above the viewer.

- **Full performance** — the viewer occupies the whole window.
- **Viewer-only window** — a second window for another display, showing the complementary view: 3D while the editor shows 2D, and the reverse.

The media panel

The media panel holds everything the project contains, including the sequences and compositions the program generates.



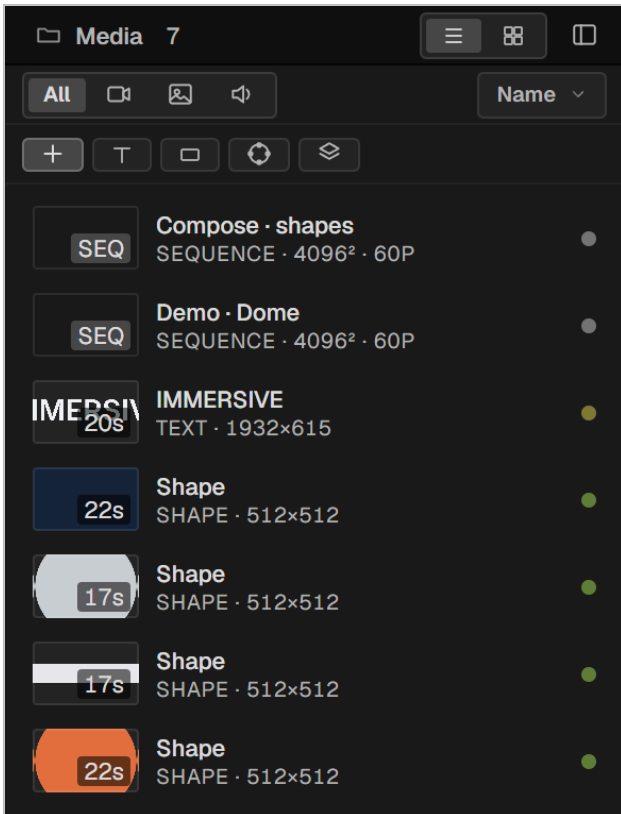
The media panel in list view.

- 1 **Kind filters** — all items, video, stills, audio.
- 2 **Sort** — by Name, Date or Type.
- 3 **View** — list or grid.
- 4 **Import** — opens the file picker; also **Ctrl** + **I**.

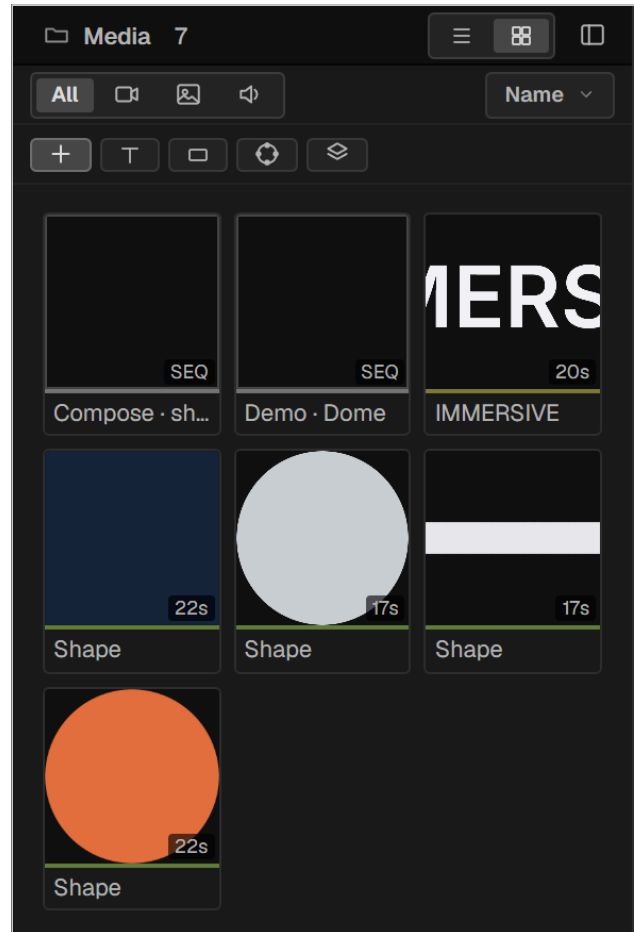
- 5 **Text** — creates a text item: font, weight, alignment, fill and outline colour, with custom fonts loadable from file.
- 6 **Compose** — builds a generated composition from the selected items. Chapter 16.
- 7 **Adjust** — creates an adjustment layer, a clip with no picture that applies its effects to everything composited below it.

Below the create row, each item shows its name, kind, resolution and duration.

Between **Text** and **Compose**, **Shape** creates a rectangle, ellipse or line with a fill and a stroke.



List view.



Grid view.

6.1 Search

Ctrl + **F** reveals a text filter in the filter row; **Esc** closes it.

6.2 Folders

Right-clicking empty panel space creates a folder. Folders nest and accept colours, and items are dragged between them. Renaming an item, moving it between folders, and creating, renaming or deleting a folder are all undoable.

6.3 Badges

A duration in brackets

The item carries in and out marks set in the source monitor and will enter the timeline trimmed.

A coloured dot

The item's colour, which its clips take on the timeline.

SEQ

The item is a sequence or a composition.

6.4 Selecting

Clicking selects, **Shift**+click extends a range, and **Ctrl**+click adds or removes one item. Multiple selection applies to generating several proxies at once and to building a composition from a set of sources.

The viewer

The viewer displays the composite master. It has a 2D mode, which shows the master as it will be delivered, and a 3D mode, which shows it mapped onto the dome or room.

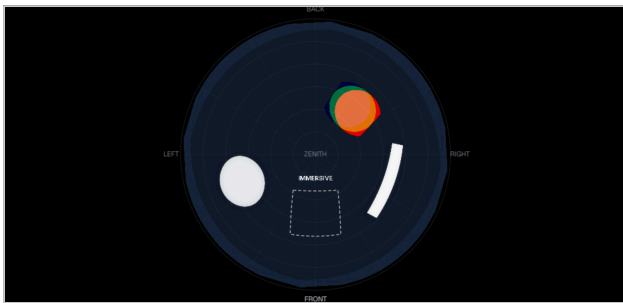


The viewer bar.

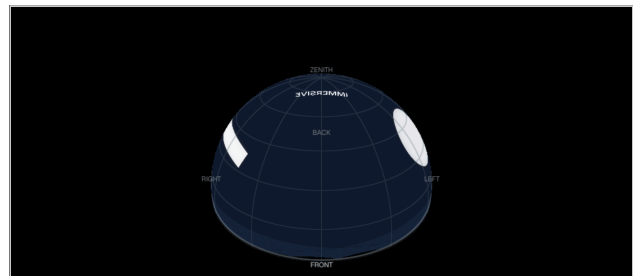
- 1 **View mode** — 2D master or 3D preview.
- 2 **Overlays** — grid, outline, horizon fade, and the floor rectangle in a room.
- 3 **Preview quality** — Full, 1/2 or 1/4. Lowering it eases playback and leaves the export untouched.
- 4 **Clip and Comp** — whether the viewer shows the selected clip alone or the whole composite, and whether composition caching is in use.
- 5 **Zoom** — the percentage, with the wheel over the viewer as an alternative. The hand tool pans.
- 6 **Output** — full performance, the viewer-only window, and the NDI and Spout broadcasts. Chapter 24.

When the window is narrow, the bar folds its less-used controls into a ... menu.

7.1 2D and 3D



2D: the dome master, with orientation guides.



3D: the same master on the dome, seen from inside.

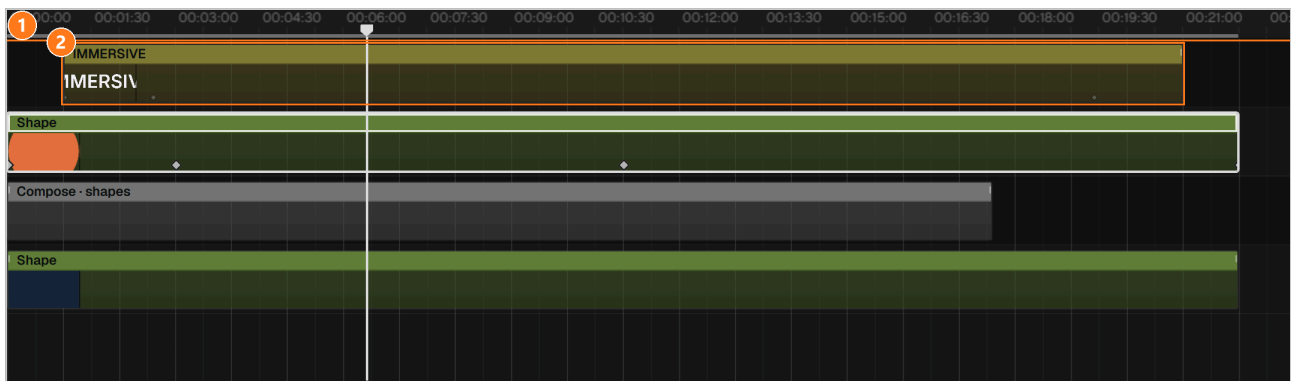
The 2D view serves for framing and delivery. The 3D view shows the shape an audience sees: a clip that reads clearly as a fisheye can be strongly distorted overhead, and the 3D view reveals that. In 3D, Orbit moves the camera freely and Viewer places it where a person stands.

7.2 Working in the viewer

Clips are dragged directly in the viewer: in a dome this moves azimuth and elevation, and in a 2D or room sequence it moves horizontal and vertical position. Point masks are drawn on the viewer canvas as well; chapter 17 describes them.

The dome circle is the picture. Material that appears in the corners of the 2D view lies outside the projected image and is recorded as black, or as transparency when the export carries alpha.

The timeline

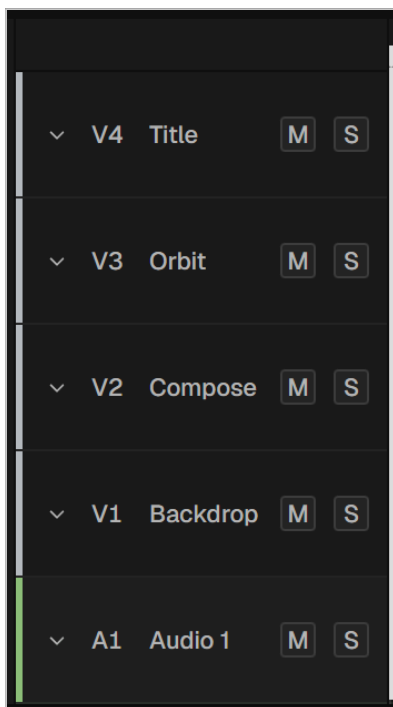


The timeline.

- 1 **Ruler** — the time scale. Dragging it scrubs the playhead, and markers occupy its lower half.
- 2 **Clip** — the diamonds along its lower edge are keyframes, and the small handles in its upper corners make fades.

Video tracks stack from the bottom up: a clip on V3 draws over a clip on V1. Audio tracks sit at the foot of the column and carry a subtle tint.

8.1 Track headers



Track headers: the tag, the name, and the mute and solo buttons.

Right-clicking a header adds, duplicates, renames, colours or deletes a track. Dragging a header up or down reorders the tracks. In a room, floor tracks carry a blue-framed tag.

8.2 Tools



- 1 **Select** (**V**) — selects, moves and trims.
- 2 **Track select** — selects everything from the click onwards.
- 3 **Hand** (**H**) — pans the view.
- 4 **Trim** (**T**) — contextual trim: ripple, roll, slip and slide.
- 5 **Razor** (**B**) — cuts a clip.
- 6 **Zoom** (**Z**) — zooms to the click.

The tool rail.

8.3 The transport



The transport bar.

- 1 **Sequence tabs** — the open sequences. Chapter 15.

- 2 **Playback** — go to start, play, go to end, and the in and out marks. **Space** plays and pauses.
- 3 **Timecode** — the playhead position, with the frame count beside it.
- 4 **Display** — timecode or frames.
- 5 **Loop** — sets the loop range from the time selection.
- 6 **Marker** — drops a named marker at the playhead; also **M**.
- 7 **Editing well** — **Auto** shows automation curves over the clips, **Grid** toggles grid snapping, **Fit** zooms to the whole project.

Shuttle

J and **L** shuttle backwards and forwards; pressing either again gives 2×, 4× and 8×. **K** stops. **↑** and **↓** jump to the previous and next cut.

With a loop range set, playback started inside the range loops, and playback started outside it runs straight through.

8.4 Zoom and navigation

Zoom	Ctrl +wheel, or + and - . Zooming is anchored to the pointer.
Track height	Alt +wheel.
Pan horizontally	Shift +wheel, or a horizontal wheel.
Scroll vertically	The wheel.

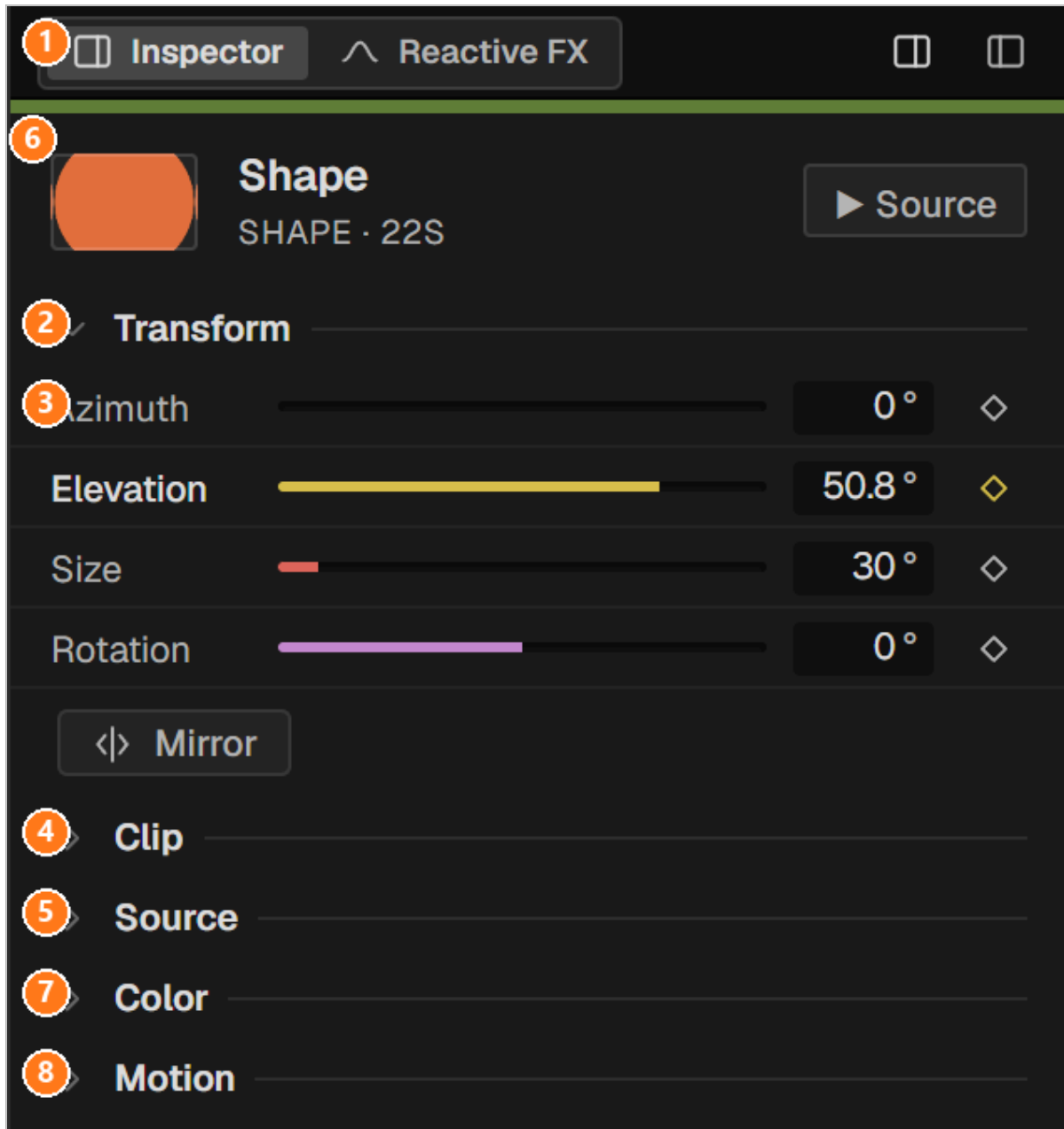
A vertical bar at the right edge mirrors the horizontal one: its body scrolls and its end caps change the track height.

8.5 Markers

M drops a marker at the playhead, and **,** and **.** jump between markers. A command in the palette detects beats in an audio clip and places markers on them.

The inspector

The inspector holds the selected clip's parameters in collapsible sections. The sections shown depend on the kind of clip: an audio clip has no Transform, an adjustment layer has no Source.



The inspector, with Transform open.

- 1 **Tabs** — **Inspector** and **Reactive FX**, the audio-driven effects chain described in chapter 20.
- 2 **Transform** — position on the dome or canvas.
- 3 **Parameter rows** — a name, a coloured track, a value box and a keyframe diamond.
- 4 **Clip** — opacity, blur, crop, mask, blend mode and black key. Chapter 17.

- 5 **Source** — how the footage is interpreted. Chapter 12.
- 6 **Playback** — speed and looping. Chapter 11.
- 7 **Color** — grading and LUTs. Chapter 18.
- 8 **Motion** — procedural movement, with the effects chain below it. Chapter 19.

9.1 Parameter rows

- Dragging the track changes the value.
- Double-clicking the value box allows an exact number to be typed.
- Clicking the diamond sets a keyframe at the playhead; clicking a filled diamond removes it.
- The track's colour identifies the parameter and is reused for its automation curve.

On a parameter that already carries keyframes, changing the value writes a new keyframe at the playhead rather than moving the whole curve. On a parameter without keyframes, the value simply changes.

9.2 Header

The top of the inspector shows the clip's thumbnail, name and kind, with a **Source** button that opens the source monitor on that item.

Importing and managing media

10.1 Import routes

Import media..., the Import button, **Ctrl**+**I**, or a double-click on empty panel space

Every image becomes its own item.

Import image sequence...

Numbered files are grouped into one sequence item, and the frame rate is requested. The question offers a third answer, **As separate images**, for numbering that is coincidental.

Dragging files onto the panel

Each file becomes its own item.

Whether numbered images form a sequence is decided by the command used, before the file picker opens.

10.2 Proxies

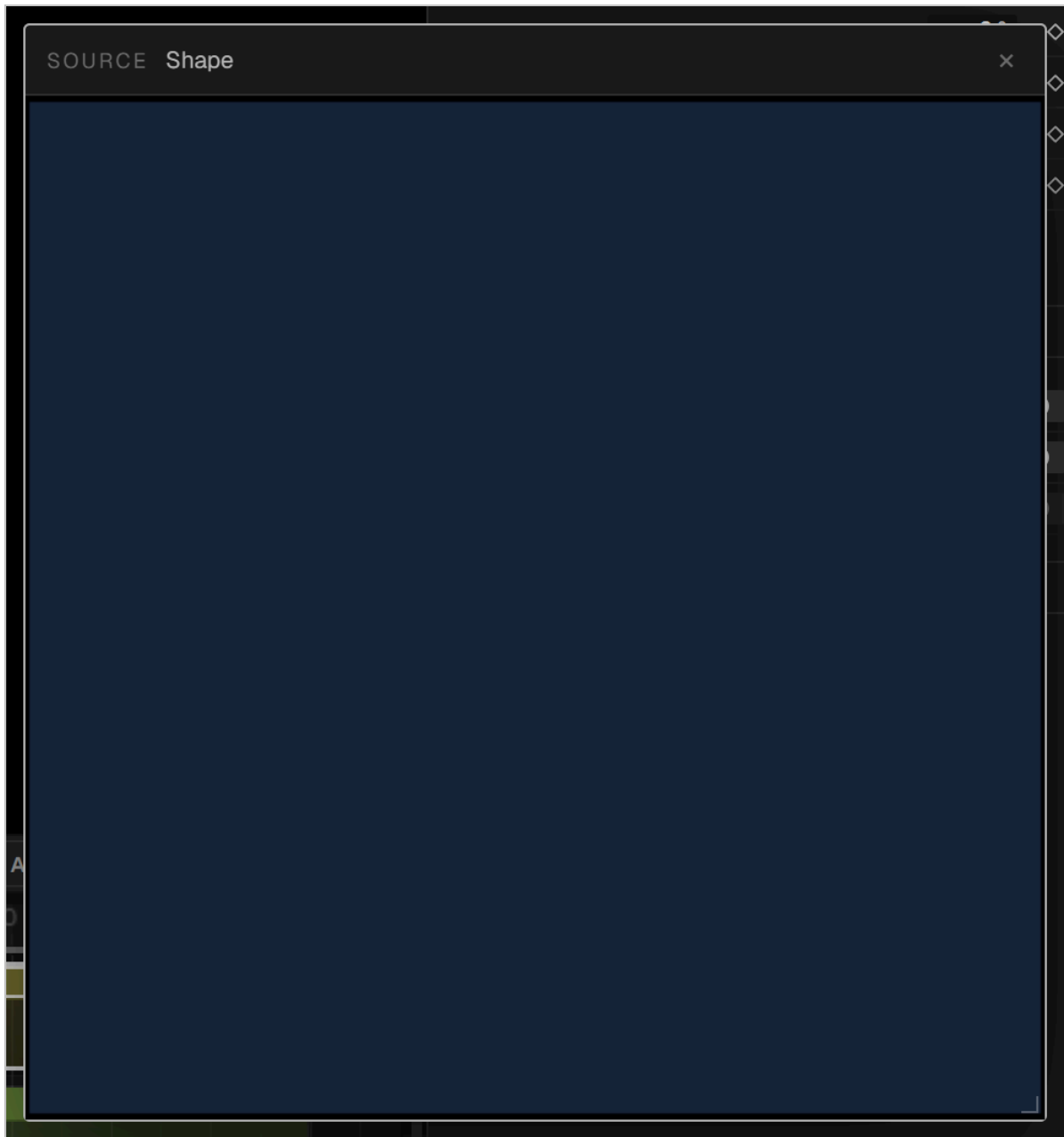
A proxy is a light, all-intra copy of a heavy clip, used while editing. Proxies are generated on request: right-click an item and choose **Generate proxy**, or select several items first to queue a batch.

- Importing heavy footage prompts a reminder in the status bar that the option exists.
- Proxies already on disk are adopted automatically, both when a project is reopened and when the same footage is imported again.
- A proxy whose generation was interrupted is detected and discarded.

With long-GOP camera masters, scrubbing a frame can take over a second. A proxy brings that to a few milliseconds.

10.3 The source monitor

Double-clicking a media item, or pressing **Source** in the inspector, opens the source monitor: a floating player for reviewing material and marking a range.



The source monitor. The picture itself is the drag handle, and only the marked range lands on the track.

- **I** and **O** set the in and out points, **Space** plays, and the arrow keys step one frame.
- The window stays open while work continues. It moves, resizes from its corner, and closes with **Esc**.
- Marks belong to the item, are saved with the project, and apply when dragging from the media panel. An item that will enter trimmed shows its duration in brackets.
- Marks change only through an explicit gesture, so reviewing material leaves them as they were.

10.4 Relinking and replacing

Relink

Points an item at a file that has moved; every clip using it follows.

Replace media

Exchanges the footage under existing clips while keeping their position, duration, effects and automation. A looping clip loops over the full duration of the new footage and keeps its length on the timeline. This applies inside nests and compositions as well.

10.5 Text and shapes

Text items carry font, weight, alignment, fill colour and outline colour, and accept custom font files. They render at a high internal resolution so they stay sharp on a 4K dome; the proportion of the canvas governs the result, so there is no pixel-size field.

10.6 Saving

Projects are .isp files in JSON, written atomically with a .bak alongside. The program autosaves, keeps a recovery history, and writes incremental versions — _v01, _v02 — from the right-click menu on the **Save** button.

Editing in the timeline

11.1 Moving clips

Dragging a clip anywhere on its body moves it, and a ghost shows the landing position. **Alt**+drag copies. Linked audio and video move together, and the linked partner is held to horizontal movement so the pair stays on matching tracks.

Overlapping

Dropping a clip on top of another trims the stationary clip, and splits it in two when the new clip lands in the middle. The source file is untouched, and no crossfade is created.

11.2 Trimming

Handles

With the select tool, dragging either end of a clip trims it.

Contextual trim, **T**

The trim tool chooses the operation from the grab point: **ripple** at an outer edge, **roll** on a cut, **slip** in the middle of a clip, **slide** below it. A linked partner is trimmed to match.

11.3 Fades and crossfades

Each clip carries a small handle in its upper corners.

- Dragging it **inwards** makes a fade in or out.
- Dragging it **outwards over a cut** makes a crossfade with the neighbouring clip. Video crossfades geometrically and audio crossfades by gain.

11.4 Cutting and deleting

Razor	B	Clicking cuts.
Split at playhead	Ctrl + E	Cuts the selection, or every clip under the playhead.
Delete	Del	Leaves a gap.
Ripple delete	Shift + Del	Closes the gap.

11.5 Copy and paste

Copying takes the whole selection. On pasting, each clip lands on its own track at its original spacing relative to the first, so a multi-layer arrangement returns intact. Linked pairs are relinked when both halves were copied.

Copy and paste attributes

Ctrl + **Alt** + **C** and **Ctrl** + **Alt** + **V** carry the properties of a clip — transform, looping, speed, colour, masks, effects, motion and all automation — to another clip. This works in both directions between an ordinary clip and a composition.

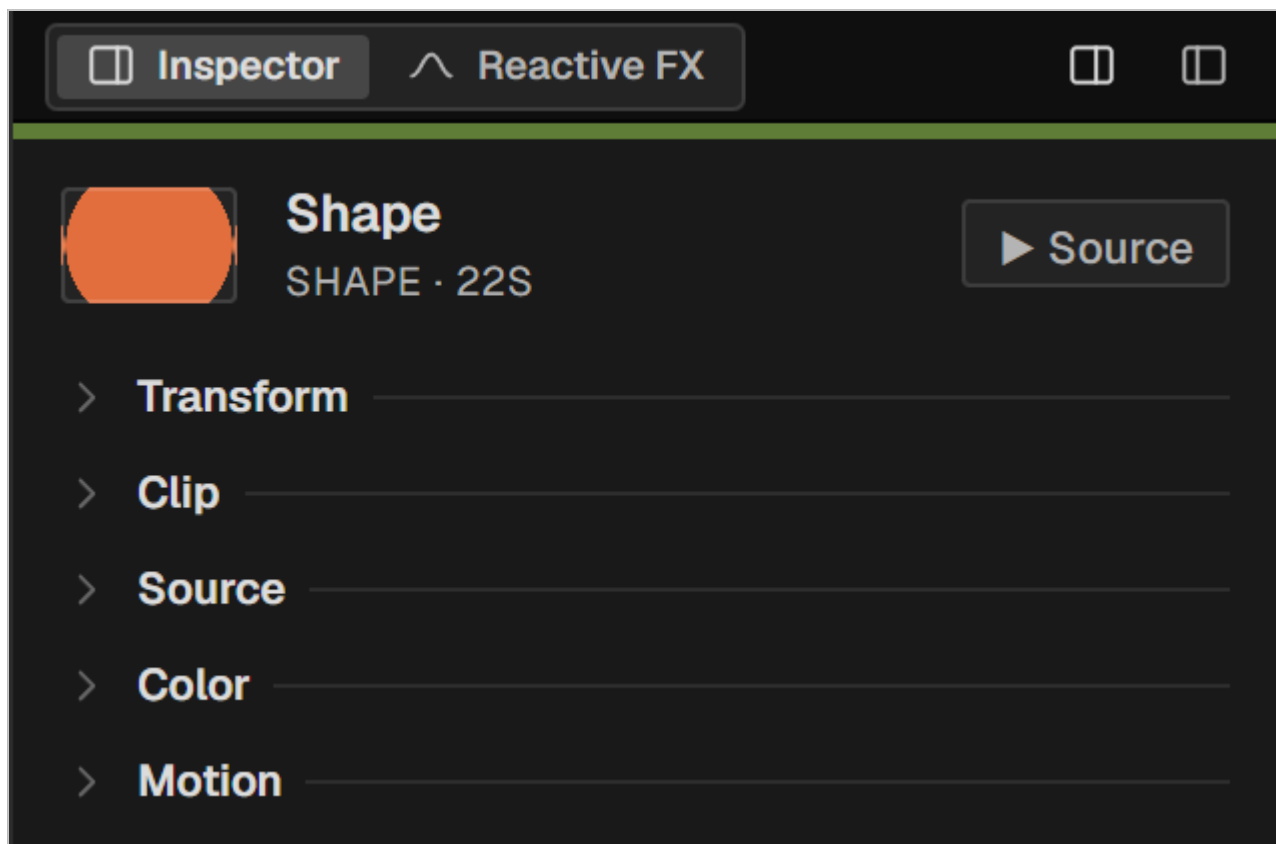
Identity, position, duration and the media itself stay with the destination, which keeps its place and its length on the timeline. A composition's generating configuration stays with its own composition.

Automation is trimmed to fit. Pasting from a 40-second clip onto a 15-second clip keeps the automation that fits, and inserts the interpolated value at the start when the first surviving keyframe falls later, so the curve begins at the right value.

11.6 Selection and snapping

- Dragging on empty timeline space makes a marquee; dragging on the ruler makes a time selection, which becomes the loop range.
- Snapping catches clip edges, the playhead, markers and the grid.
- **Ctrl** + **D** duplicates, and **Alt** + **←** and **Alt** + **→** nudge by one frame.

11.7 Speed and looping



The Playback section.

Speed

50 % to 200 %. Changing it stretches the clip together with its automation, so keyed movement stays in the same place relative to the picture.

Loop

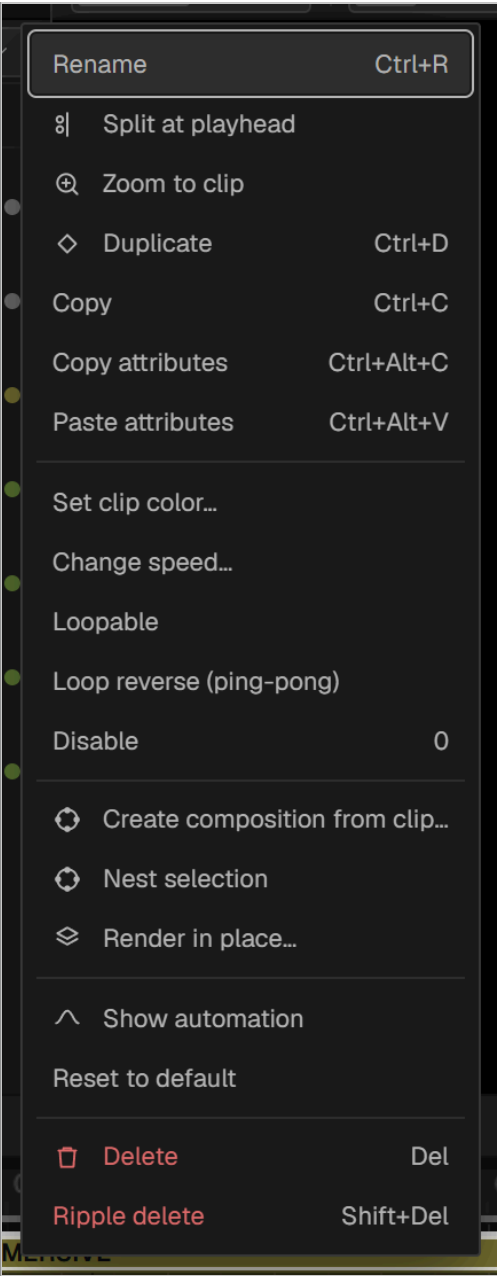
Repeats the source for the length of the clip. The inspector shows the cycle length and the part of the source it repeats, in timecode.

From clip

Reads the cycle length from the clip's current duration.

Switching looping off trims the clip to the remaining length of the file. To redefine a cycle, use **From clip**, which changes the cycle length alone.

11.8 The clip menu



Right-clicking a clip.

Placing clips in the dome

12.1 Transform

Azimuth	Bearing around the horizon, in degrees. 0° faces the front of the dome.
Elevation	Height above the horizon, in degrees. 90° is the zenith. Negative values place a clip below the horizon, which is projected area in a dome with coverage above 180°.
Size	How much of the dome the clip subtends.
Rotation	Roll about the clip's own centre.
Mirror	Flips the clip horizontally.

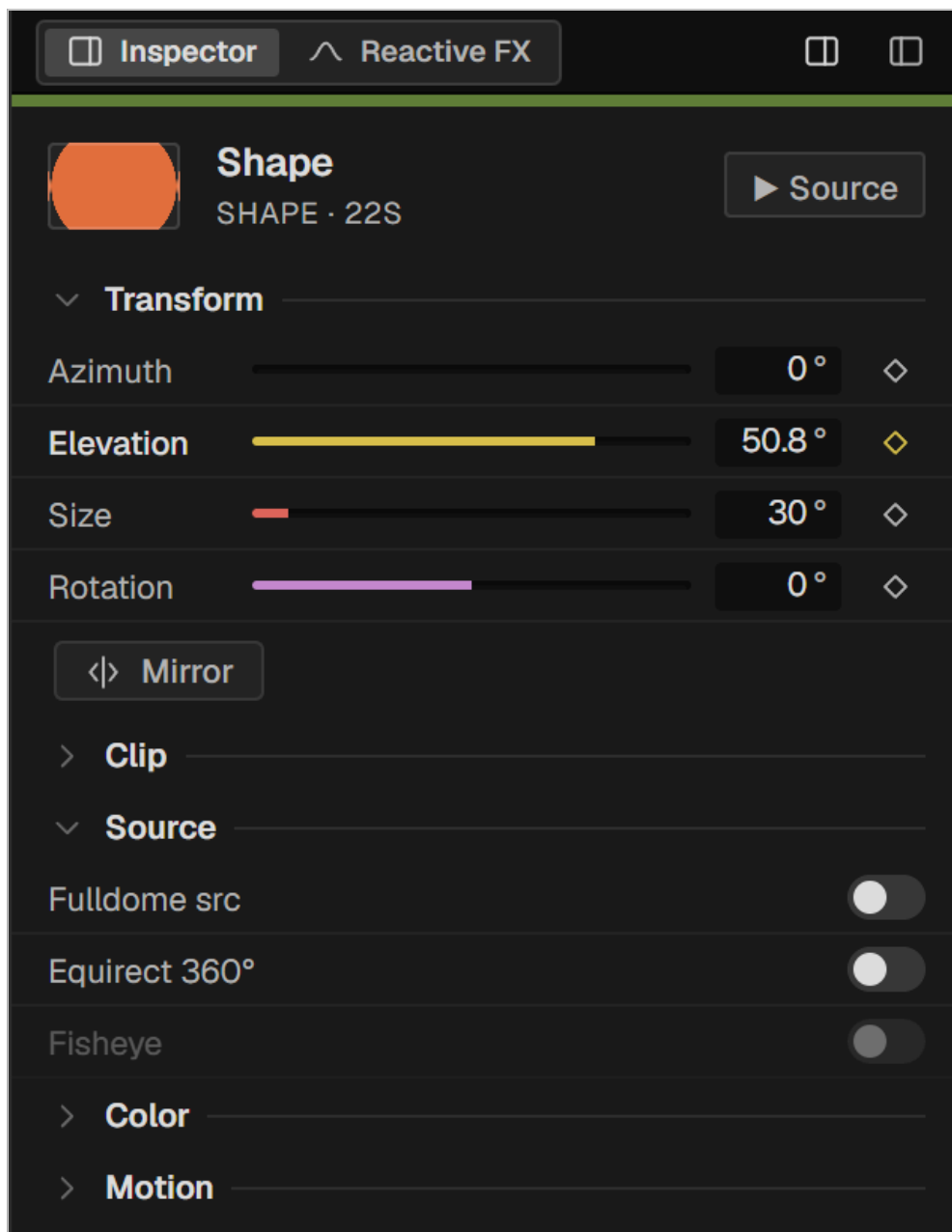
Each value can be dragged, typed, keyframed and animated. Dragging the clip in the viewer moves azimuth and elevation together.

12.2 Shape on a curved surface

A rectangle placed on a dome becomes a curved quadrilateral, and the effect grows with size and with height. Two consequences follow.

- A large clip near the zenith stretches noticeably, which the 3D view shows.
- Near the zenith one degree of azimuth covers a short distance. For this reason the **Float** motion preset animates in the dome plane rather than in azimuth and elevation, so the same drift reads the same wherever the clip sits.

12.3 Source



Transform and Source in the inspector.

Fulldome src

For footage that is a dome master — a circular fisheye filling the frame. It is drawn one to one.

Equirect 360°

For equirectangular 360° footage, reprojected into the dome.

Fisheye and Amount

Bends a fulldome source further. The row becomes available once **Fulldome src** is on, with a tooltip explaining the dependency. Switching **Fulldome src** off also switches this off.

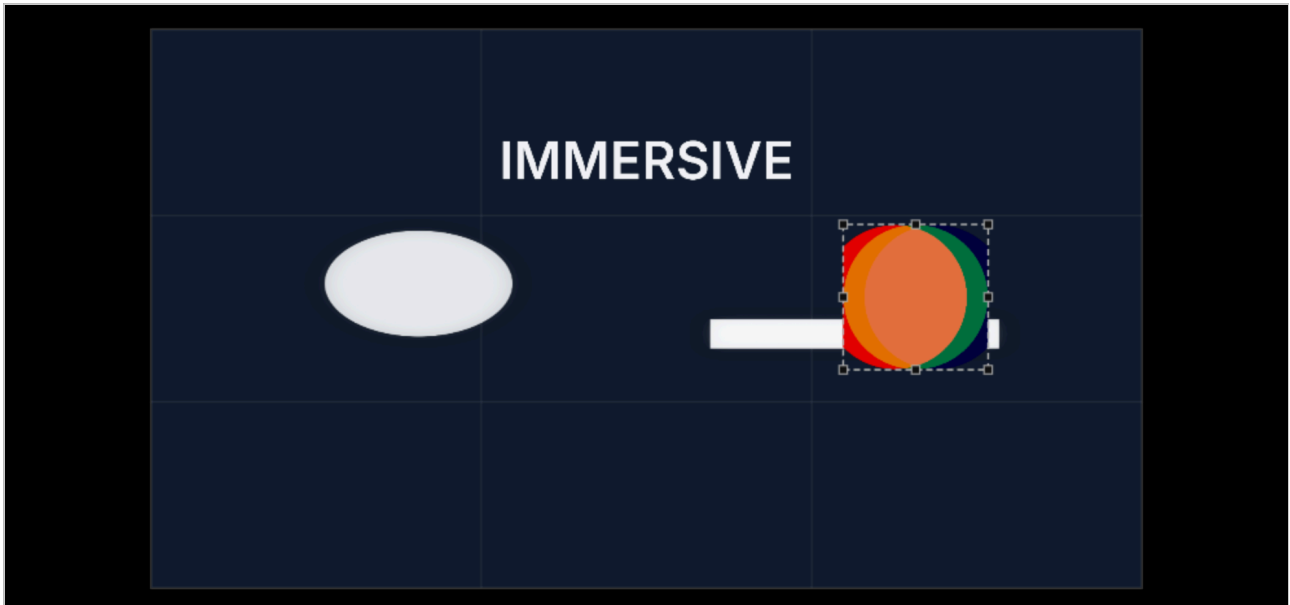
A generated composition covers the whole dome, so its clip arrives with **Fulldome src** already on. Placement happens inside it, on the elements it contains.

12.4 Adjustment layers

An adjustment layer is a clip with no picture of its own. It applies its effects chain across the full dome to everything composited below it, and carries the complete effects catalogue.

2D sequences

A 2D sequence uses a rectangular canvas. The rest of the program behaves as it does in a dome.



A 2D sequence, with the safe grid drawn over the canvas.

13.1 Differences

Transform	X, Y, Scale and Rotation.
Source	The dome switches apply to dome sequences.
Viewer	The 2D master fills the viewer.
Motion presets	Rotate, Pulse, Horizontal, Vertical and Float.
Compose	The flat layouts: grid, line, row, column and weave.

13.2 2D inside dome work

Two parts of the program use flat sequences directly.

- **Compositions are built flat.** The Weave composition lives in a 1:1 flat nest, where neighbouring strips meet at right angles, and enters the dome as a fulldome source with fisheye applied to the whole assembly at once.
- **The floor of a 360° room** is edited as a flat area of the room canvas.

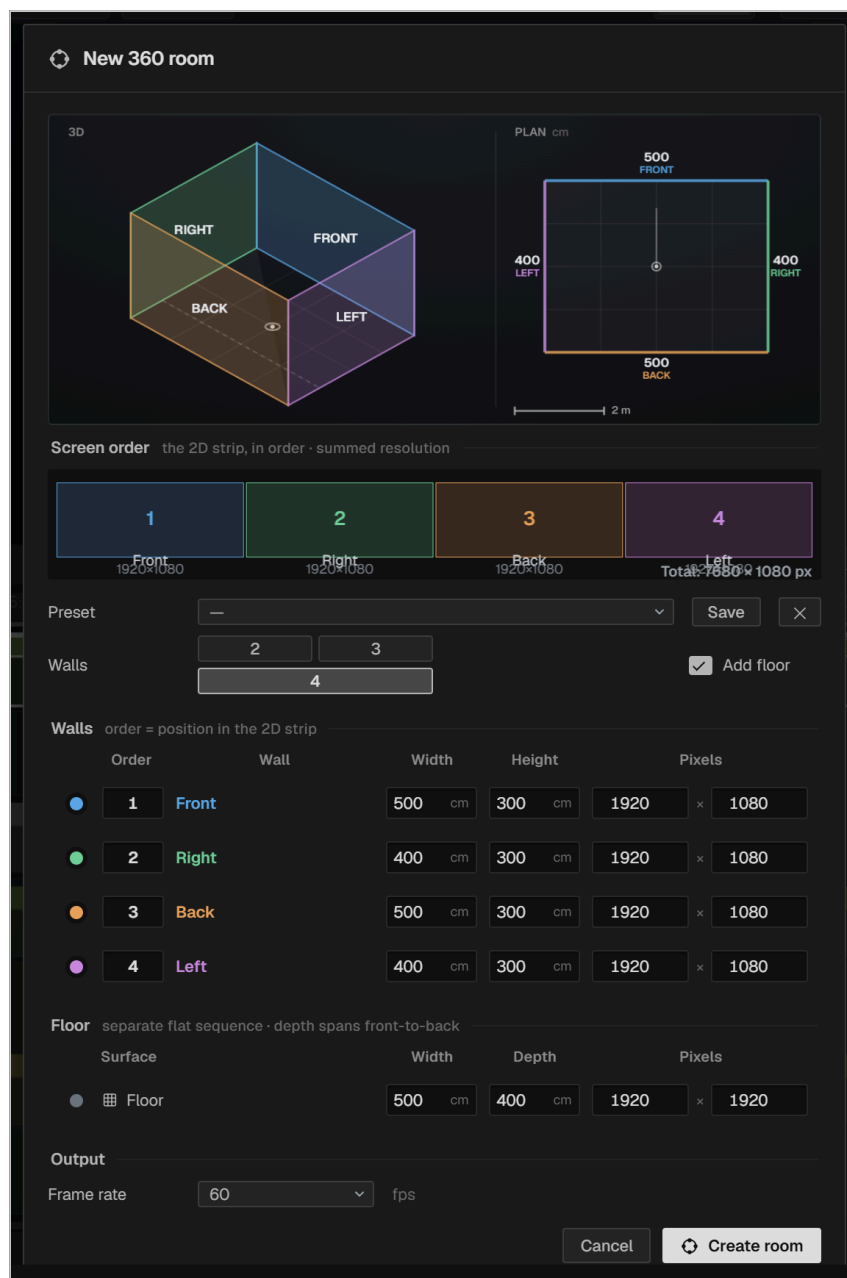
Project → New sequence... adds a 2D sequence to a project of any format.

The 360° room

A room treats several projection surfaces as one piece: walls unrolled into a single wide strip, an optional floor, a 3D view from the audience position, and an export that delivers the whole strip or one projector's area at a time.

14.1 Geometry

The room is defined when the project is created, and reconfigured from **Project → Room geometry....**

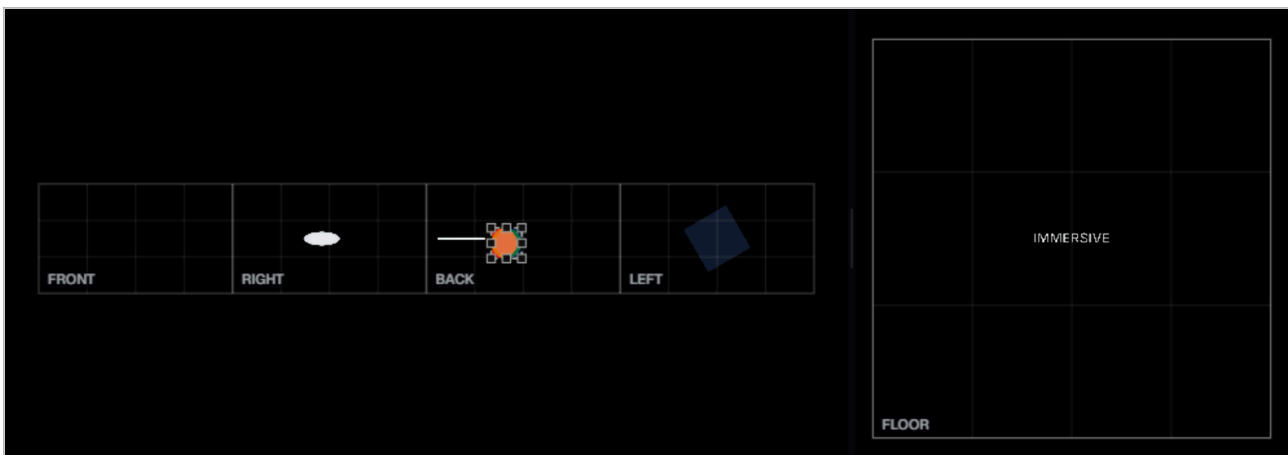


Room setup. The plan view shows whether the walls close into a room.

Each wall carries a role — front, right, back, left — a physical width and height in centimetres, and a pixel resolution. The plan preview solves the wall loop into a floor plan and reports geometry that fails to close, or that closes by crossing itself.

Centimetres describe the room; pixels describe the canvas. The strip is measured in pixels, and its width is the sum of the walls' pixel widths. The physical measurements give the 3D view its proportions and derive the floor.

14.2 The strip

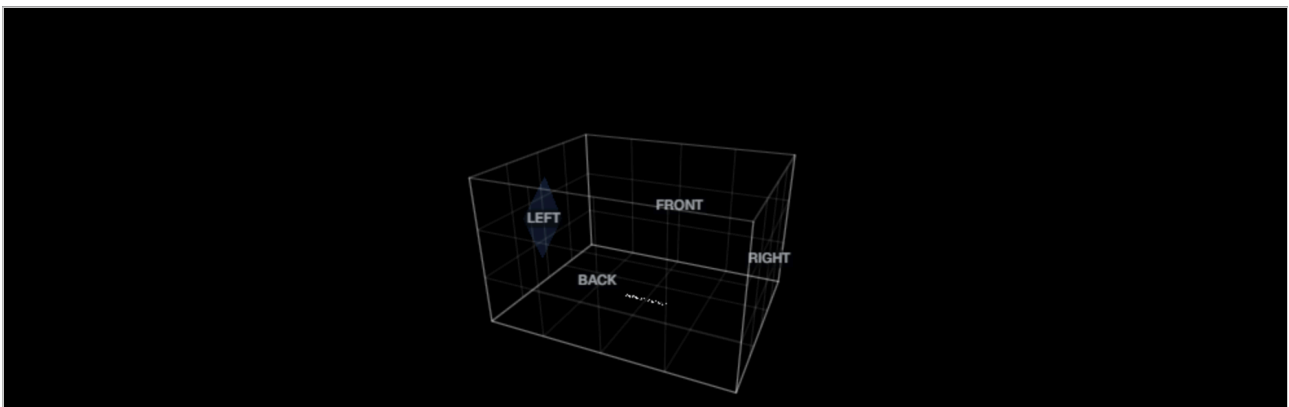


The unrolled strip, with seams and wall labels drawn over it.

The strip is edited as a 2D canvas: clips take a horizontal position, a vertical position and a scale, and a clip can span a seam and run across two walls. The seams and labels are guides.

Floor content goes on floor tracks, marked by a blue-framed tag in the track header. The floor is a band of the same canvas, so one composite covers every surface.

14.3 The 3D view

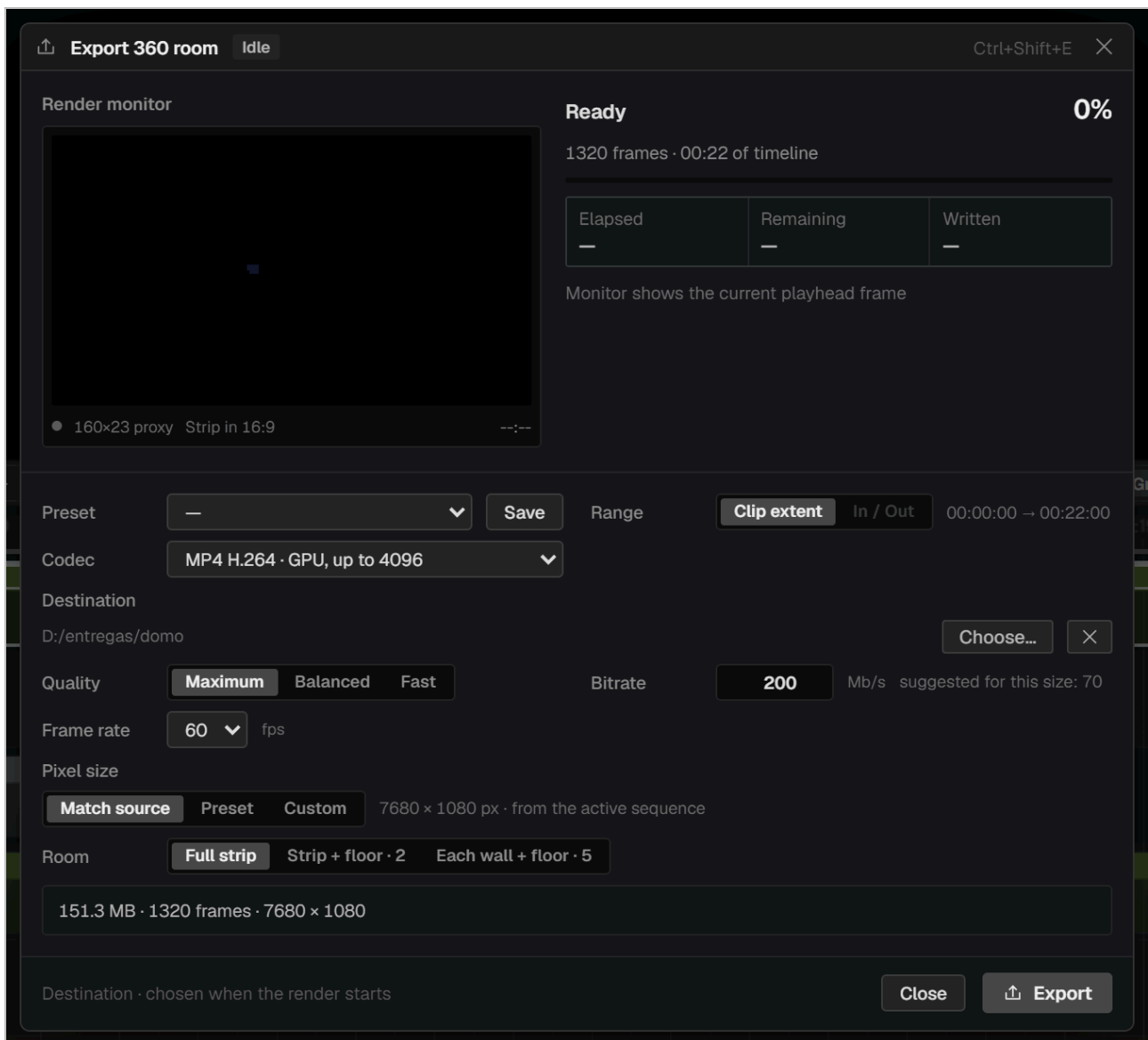


The room in 3D, with the camera orbiting or standing where a person stands.

The content is drawn unshaded, since the view previews what will be projected. The shading visible on the outside belongs to the translucent shell around the room.

14.4 Exporting a room

Full strip	The walls as one wide file.
Strip + floor	Two files: the wall strip and the floor.
Each wall + floor	One file per wall plus the floor, one per projector. Available when the room has a floor.

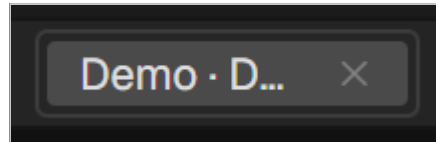


Exporting a room. The mode adds its jobs to the queue.

Sequences and nests

A sequence is a media item, and a media item can be placed on a track. Sequence tabs, nesting and generated compositions all follow from this.

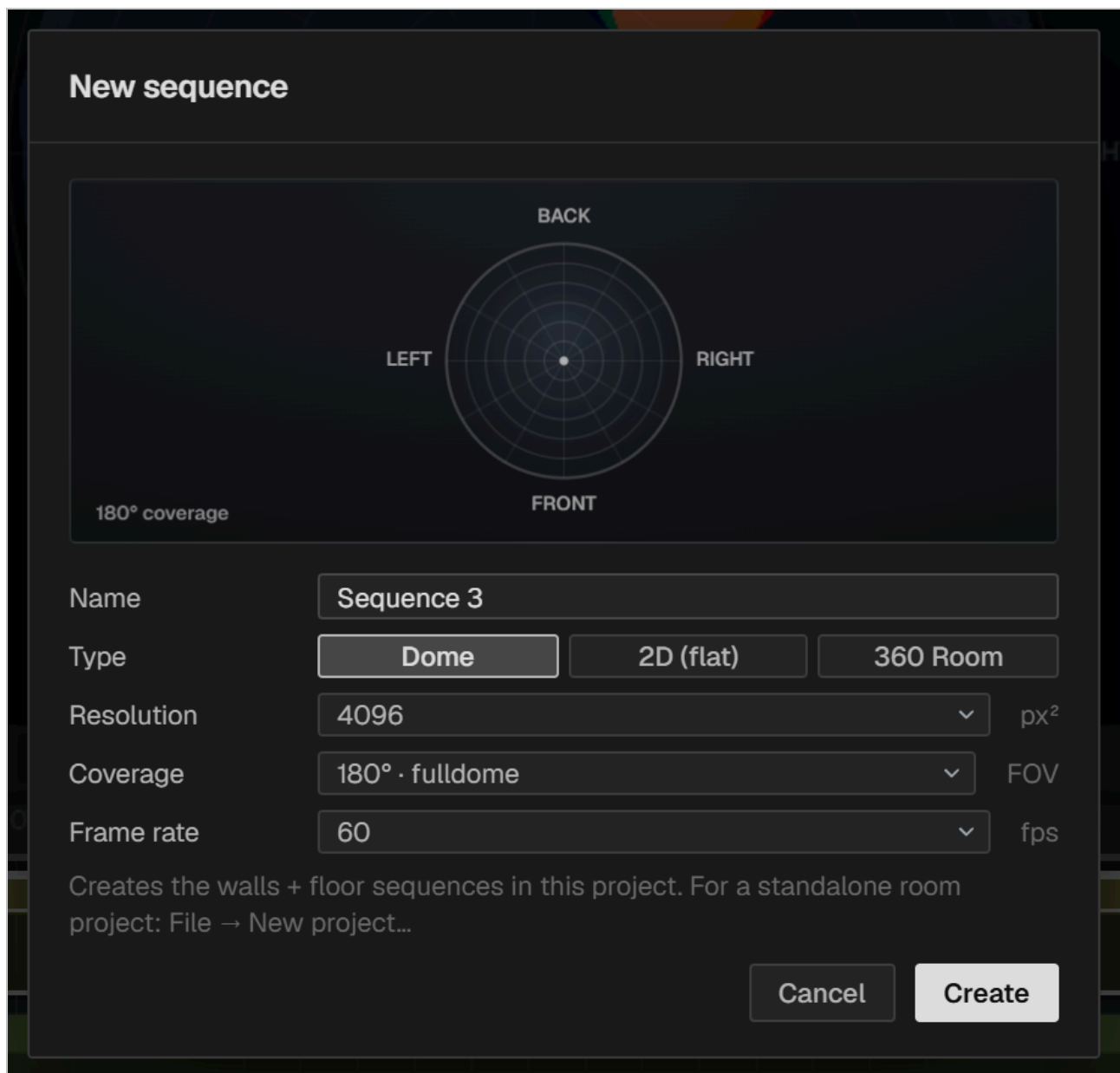
15.1 Tabs



Sequence tabs, at the left of the transport bar.

Tabs hold a fixed width so the bar stays still as sequences are renamed. Three are visible at a time and the rest are reached with the wheel over the tabs. Dragging a tab reorders it, and + creates a sequence.

15.2 Creating a sequence



New sequence: dome, 2D or 360 Room, each with its parameters and a preview.

15.3 Nesting

Selecting clips and choosing **Nest selection** from the **Edit** menu, or from the clip menu, replaces them with a single clip containing them. Double-clicking it opens its contents, and the tab bar shows the current position.

- A nested clip holds an independent copy, so editing it leaves the originals as they were.
- In a dome sequence a nest is created as a fulldome source: it composites internally and enters the dome as a complete dome image.
- Shortening the contents of a nest shortens its instances in the parent sequence.

Audio inside a nest

Sound plays from audio tracks that are visible. A nest containing audio tracks therefore creates a derived audio clip in the parent, linked to the video clip, which moves, trims and unlinks like any A/V pair.

15.4 Composition proxies

A nest can be cached as a single video, so playback decodes one stream instead of many. The cache renews itself when the contents change.

15.5 Render in place

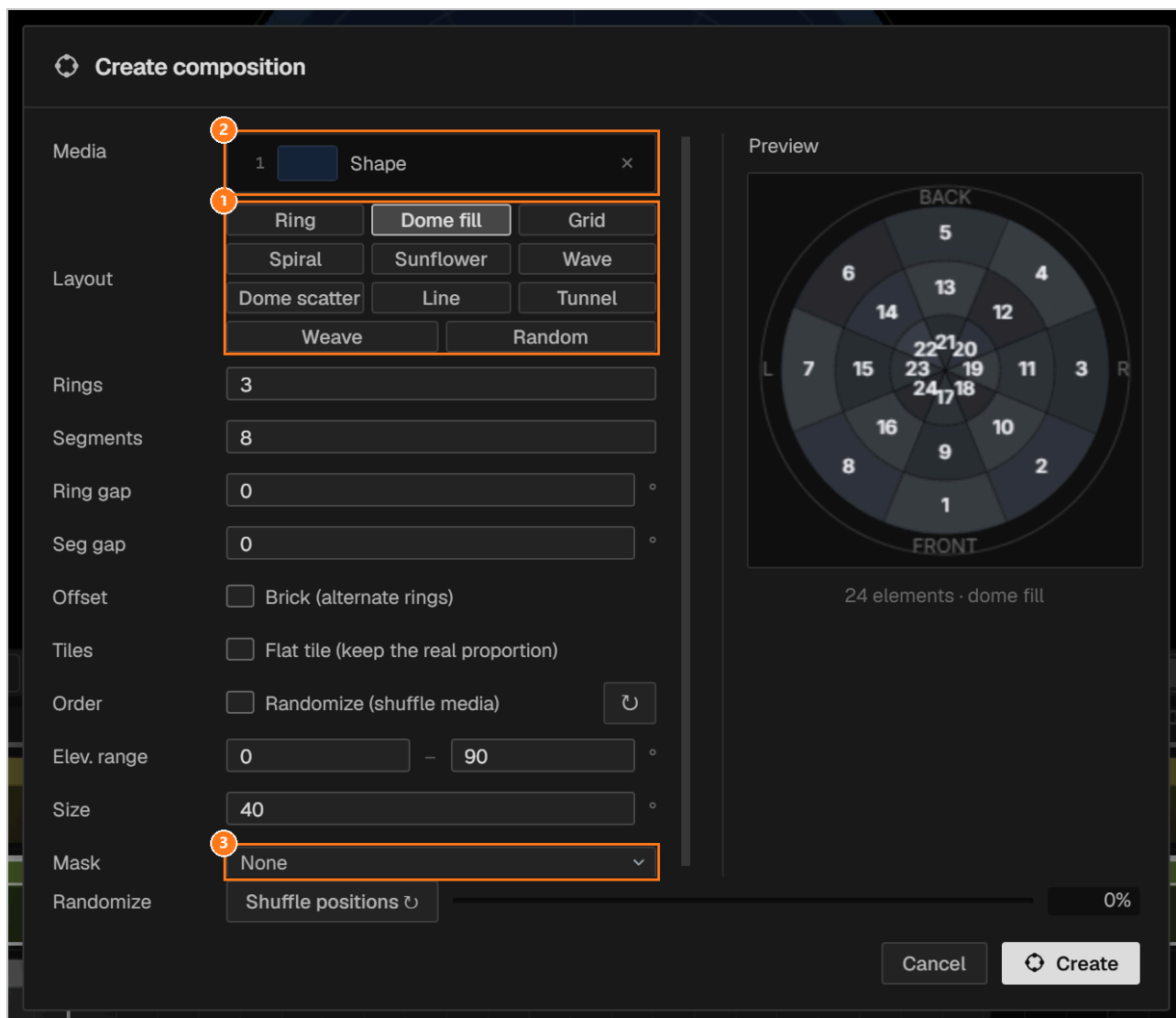
Render in place... in the clip menu bakes a clip or a nest to a silent MP4 and replaces the timeline instance with the baked file on a new track. A progress view shows the live frame, a bar and an estimate, and offers cancellation.

Compose

Compose builds an arrangement out of media items — a ring of clips around the horizon, a tunnel approaching the viewer, woven strips filling the dome — and returns it as an editable nested sequence.

16.1 Building a composition

1. Select the items to use in the media panel.
2. Click **Compose** in the create row.
3. Choose a type and set its parameters. The preview follows each change.
4. Confirm. The composition lands on the nearest track with room for it.



The Compose dialog. The editor remains visible behind it, so the composition can be judged against what is already on the dome.

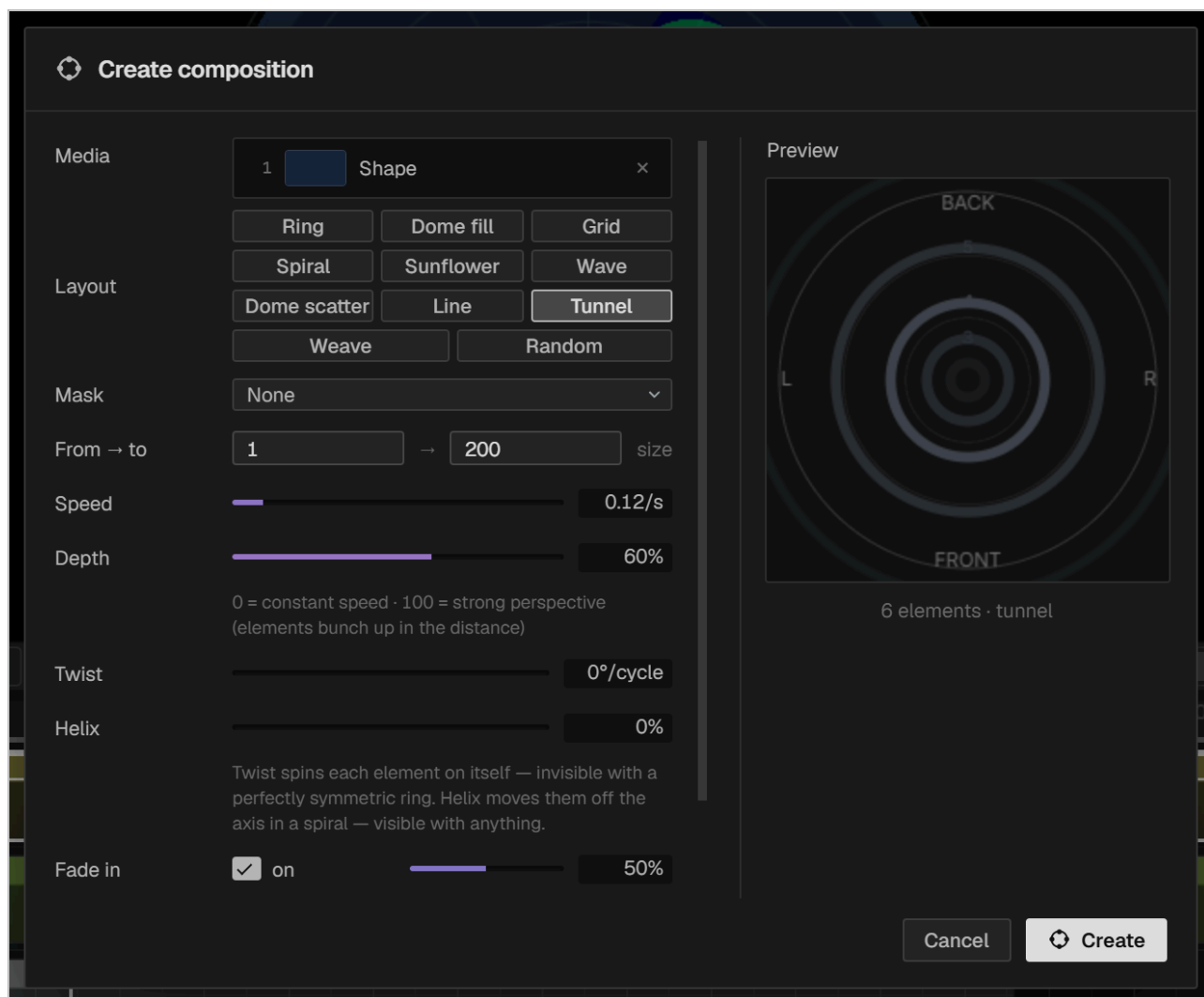
- 1 **Type** — the layout to generate.
- 2 **Media basket** — the items the composition contains, with thumbnails, names and order. × removes one; dragging from the media panel, or double-clicking there, adds one. The order decides which source goes where.
- 3 **Mask** — a shape and a size from 0 % to 100 %, applied to every element. This turns square footage into a ring of circles in one step.

16.2 Types

TYPE	RESULT
Ring	A circle of clips at one elevation.
Dome fill	A grid of rings and segments covering the dome. Tiles keep their proportions.
Grid	A rectangular grid.

Spiral	A spiral from the zenith outwards.
Sunflower	Phyllotactic packing: even coverage with no visible rows.
Wave	A wave-shaped band.
Dome scatter	Even quasi-random scattering over the hemisphere.
Line	A straight line of clips.
Tunnel	Elements grow from the centre and pass the viewer.
Weave	Interlaced strips travelling endlessly.
Random	Random placement with a jitter control.

16.3 Tunnel



Tunnel.

Tunnel sources are 1:1 images with alpha, marked as fulldome, that start small and grow until they leave past the edge of the dome. Any distribution of alpha works; a ring reads as a legible tunnel.

Each element is phase-shifted, which makes the stream continuous.

From and To

The size range each element travels through.

Speed

The rate of travel.

Depth

Bends the growth curve. An exponential rise reads as perspective, since approaching a viewer at constant speed multiplies the radius.

Spin

Degrees of rotation per cycle. A perfectly symmetrical source hides it.

Helix

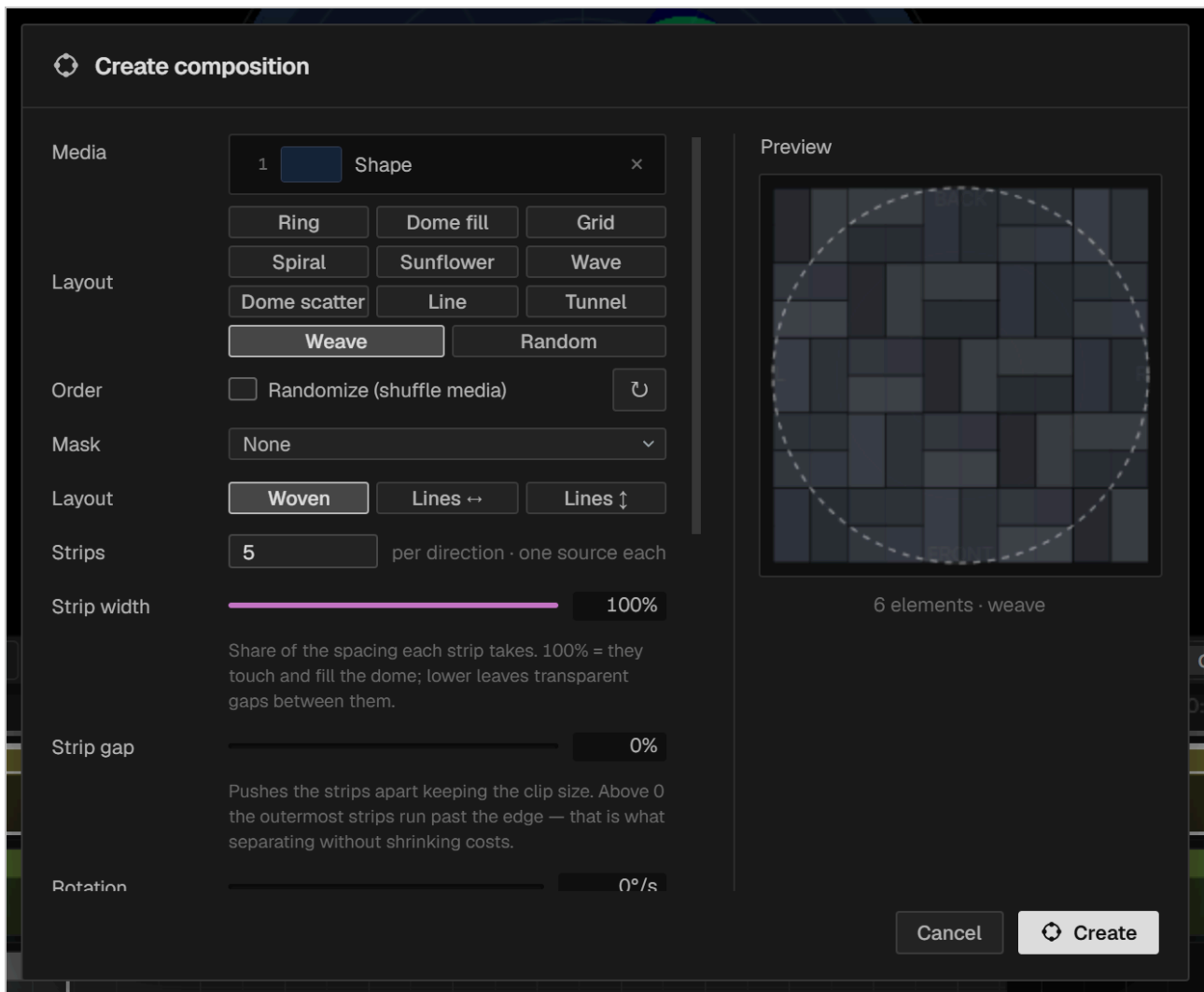
Pushes elements off the axis in a spiral, visible with any source.

Fade in and Fade out

Separate envelopes, each with an amount as a percentage of the cycle.

Elements are drawn in depth order every frame, since their ranking by nearness rotates as they cycle.

16.4 Weave



Weave, built in a flat 1:1 nest and brought into the dome as one fulldome source.

Weave keeps its sources undistorted: the side that crosses a strip takes the strip's thickness, and the other follows the media's proportions. Each strip uses one source. The interlacing is decided per crossing, so the pattern stays still while the strips flow through it.

Layout

Woven, Lines ↔ or Lines ↑.

Strips

1 to 24 per direction, one source each.

Strip width

A percentage of the step. At 100 % strips touch and fill the dome; below that a transparent gap opens. Width and spacing are one control.

Packing

Reaches 100, where one clip begins as the previous ends.

Direction

One way, The other, Alternate or Still.

Order

Shuffles which source goes to which strip.

16.5 Editing a composition

Two surfaces edit the same values.

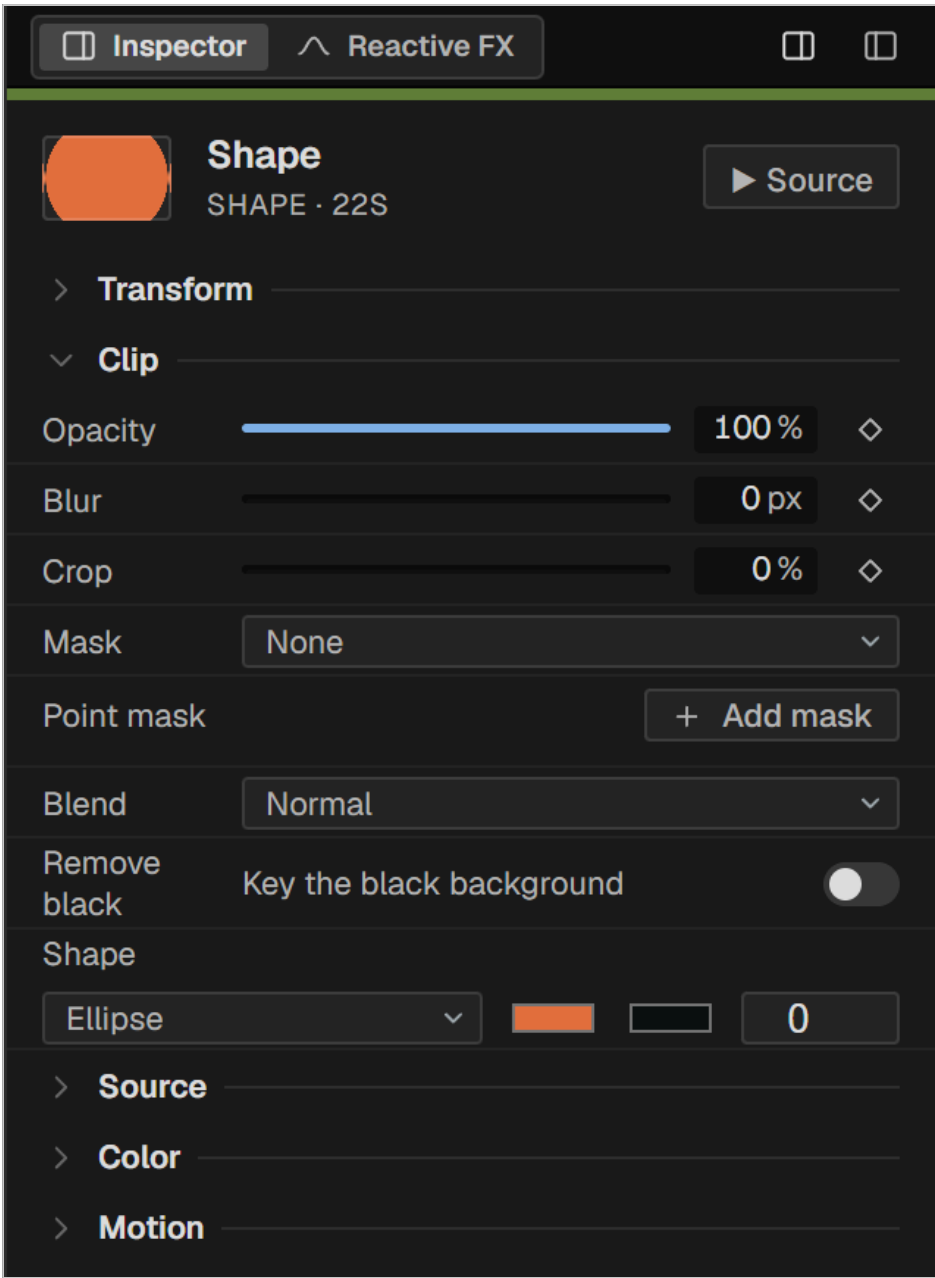
- **The inspector** shows the type, the two most-used parameters for that type, and **More options....**
- **The dialog** reopens the full parameter set.

The composition is a nest, so its contents can also be opened and moved by hand. Manual changes survive regeneration.

A composition lands on the nearest track with a free gap for its duration, measured from the selected track header, or from the bottom when no track is selected. A new track appears only when no existing track has room.

Effects and masks

17.1 The Clip section



The Clip section.

Opacity	0 % to 100 %.
Blur	In pixels.
Crop	Trims the edges of the source.
Mask	A shape or an image mask.

Point mask	Add mask begins a drawn mask.
Blend	Normal, Add, Screen, Multiply, Darken, Lighten.
Remove black	Keys out a black background, for footage delivered without alpha.

17.2 Masks

Shape masks

The **Mask** menu offers **Circle**, **Rounded**, **Diamond** and **Vignette**, and **Custom** takes a PNG as a matte. A size control scales the mask and a feather softens its edge.

Point masks

Add mask draws a mask by clicking points directly on the viewer canvas. A clip can carry several, each with its own invert and feather, which isolates part of a shot rather than cutting it to a geometric shape.

17.3 The effects chain

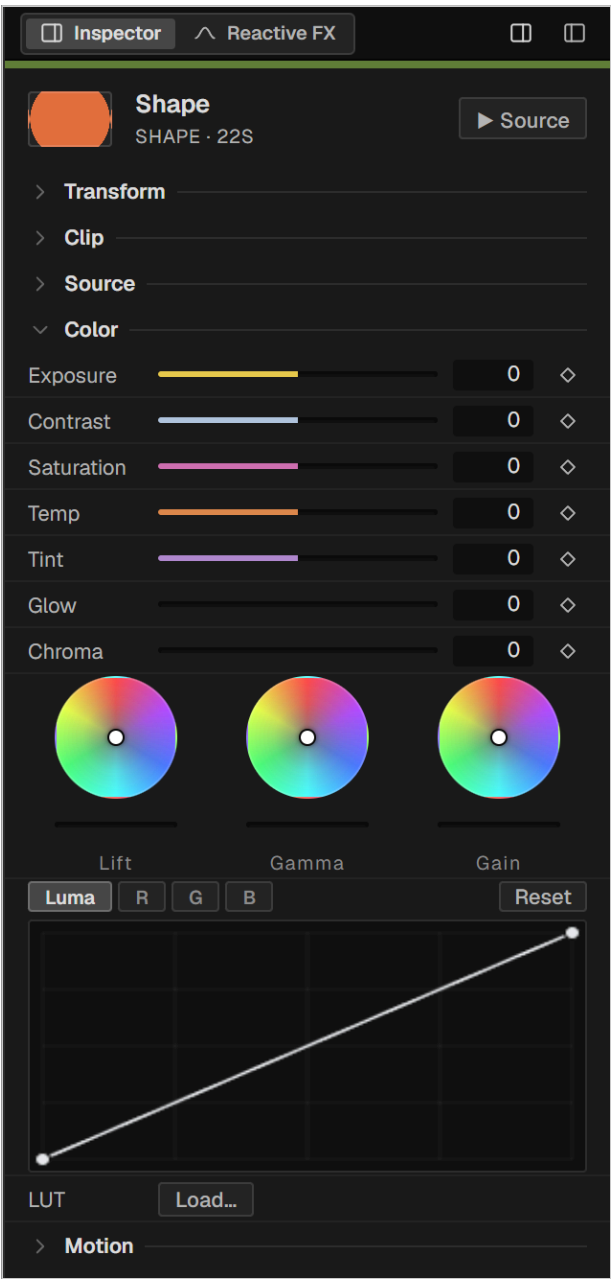
Effects appear in the **Effects** section of the inspector and in the **Reactive FX** tab, which show the same chain. **Add Effect** adds one, dragging reorders, the power button switches one off, and the bin removes it.

Every effect parameter accepts keyframes, and every effect accepts audio drive. Chapter 20 covers the audio side and chapter 28 lists all 26 effects with their parameters.

To treat the whole picture rather than one clip, use an adjustment layer, created from **Adjust** in the media panel or from **Add Adjustment Layer** in the Reactive FX tab.

Colour

Colour is set per clip and runs in the shader, so it applies identically in the viewer, the live outputs and the export.



The Colour section: sliders, three-way wheels, tone curves and a LUT slot.

18.1 Sliders

Exposure Overall brightness.

Contrast

Saturation	
Temp	Warm to cool.
Tint	Green to magenta.
Glow	A soft bloom around highlights.
Chroma	Chromatic aberration.

Each slider accepts keyframes.

18.2 Lift, Gamma, Gain

The three wheels form a primary grade: **Lift** for shadows, **Gamma** for midtones, **Gain** for highlights. Dragging inside a wheel pushes the colour, and the bar beneath it sets the level.

18.3 Curves

A tone curve with **Luma**, **R**, **G** and **B** channels. Clicking adds a point, dragging shapes the curve, and **Reset** straightens it.

18.4 LUTs

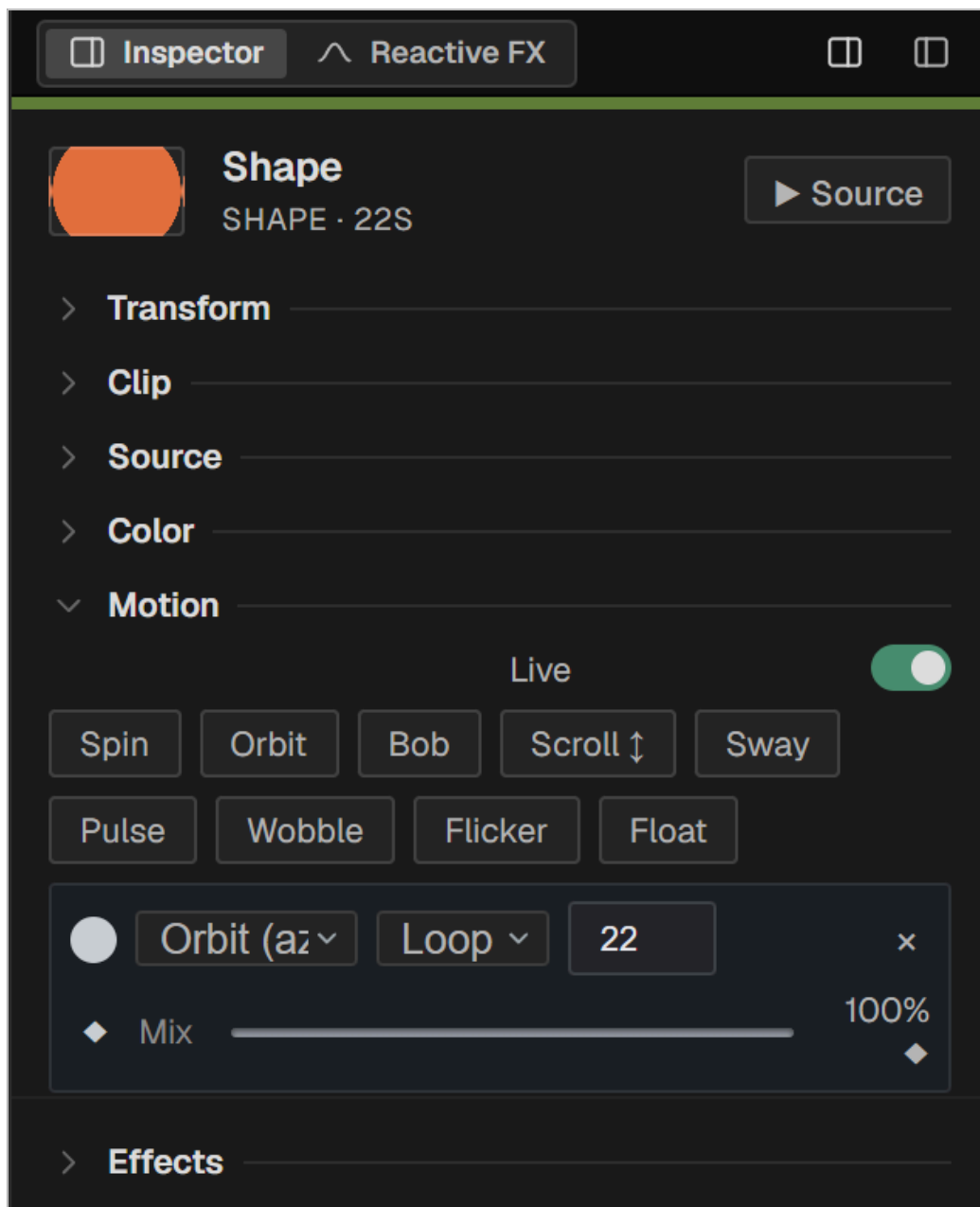
LUT → **Load...** imports a **.cube** 3D LUT and applies it per clip as the final step of the colour chain. A look developed in a grading application travels this way.

A dome is a large, close, curved surface. Material judged on a monitor often reads brighter in the venue, so exposure decisions hold better when made on the dome itself, or at least in the 3D view.

Motion, keyframes and automation

Movement comes from two mechanisms. **Motion** is procedural: endless, parameterised, with no keyframes. **Automation** is keyframed: exact values at exact times. They stack, and a mix control balances them.

19.1 Motion



The Motion section: preset chips, then the modifiers with their parameter, waveform, speed and mix.

Clicking a preset chip adds a modifier. A preset stamps ordinary modifiers into the list, which are then retuned, switched off or deleted individually.

PRESET (DOME)	PARAMETER
Spin	Rotation about the clip's centre.
Orbit	Azimuth.
Bob	Elevation, back and forth.
Scroll ↑	Elevation, continuously.
Sway	Azimuth, back and forth.
Pulse	Size.
Wobble	Roll.
Flicker	Opacity.
Float	A chord of three modifiers.

A 2D or room sequence offers **Rotate**, **Pulse**, **Horizontal**, **Vertical** and **Float**.

Waveforms

Loop

Grows without end, for continuous travel.

Wave

Goes and returns.

Saw

Rises to the amplitude and restarts at zero, for a cycle that repeats: an endless scroll, a tunnel.
Its **Curve** control from 0 to 100 bends the rise, and an exponential rise reads as perspective.

Float

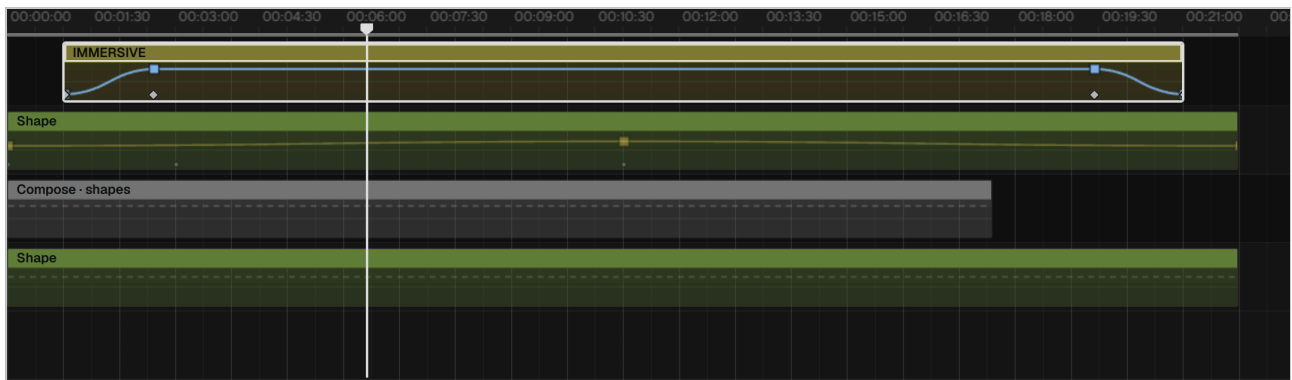
Float is a chord rather than a single parameter: horizontal drift, vertical drift and a gentle roll, out of phase with one another. Its three modifiers carry periods of 11, 14 and 18 seconds, which share no common multiple, so the combined path does not repeat. A master **Intensity** control from 0 % to 300 % scales the chord and preserves values retuned by hand.

In a dome, Float uses the slide parameters rather than azimuth and elevation, since one degree of azimuth covers less ground near the zenith than at the horizon.

19.2 Keyframes

Every parameter row carries a diamond, which sets a keyframe at the playhead and removes it when already filled. Once a parameter carries a keyframe, editing its value writes another one at the playhead.

19.3 The automation lane



Automation mode, switched on with **Auto** in the editing well. Curves are drawn over the clips in the parameter's own colour.

Each track shows one parameter's curve, chosen by two chips in the track header: the left chip picks the category or device — **Transform**, **Clip**, **Color**, or any Motion or Effect in use — and the right chip picks the parameter. A ♦ marks what is already automated. Touching a parameter in the inspector brings its curve into view.

Working on a curve

- Clicking adds a point, dragging moves it, right-clicking types an exact value.
- Dragging a selection over several points allows them to be nudged, scaled or tapered.
- The shape box applies a free transform to a selection.
- Easing presets apply standard cubic-bezier shapes to a segment.
- Simplify thins a dense curve and keeps its shape.
- A curve can be copied and pasted at the click position.

19.4 Mixing motion and automation

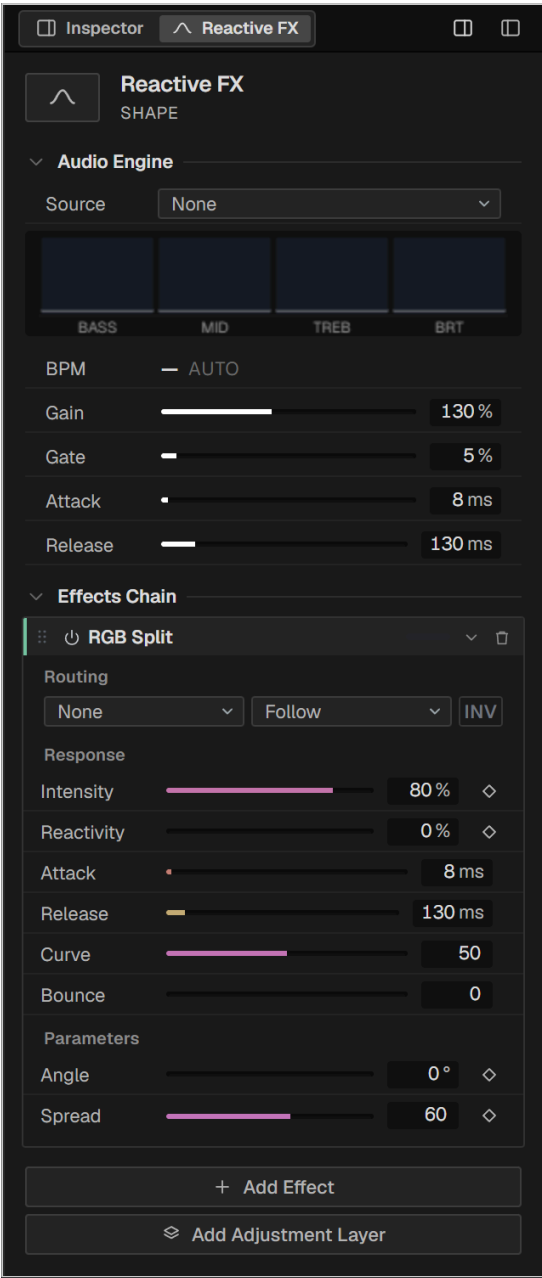
Each Motion modifier carries a **Mix** from 0 % to 100 %, and the mix accepts keyframes. A procedural drift can therefore fade in and out across a scene while keyframes handle the composition underneath.

19.5 Modulation

A parameter can also be driven by a stack of modulation layers — LFOs, audio followers and spatial functions — from the modulation panel, which suits a value that wanders rather than arrives.

Reactive FX

The second inspector tab holds an audio analysis engine and the effects chain its output can drive.



The Reactive FX tab: audio engine above, effects chain below.

20.1 The audio engine

Source	The audio that drives the analysis.
Band meters	BASS, MID, TREB and BRT: live levels for three frequency bands and overall brightness.

BPM	Detected tempo, or a value set by hand.
Gain	Input level into the analysis.
Gate	The threshold below which the picture holds still.
Attack / Release	How quickly the follower rises and falls, in milliseconds. These are the engine defaults, which each effect can override.

20.2 An effect in the chain

Each entry carries a power button, a name, a collapse arrow and a bin, and expands into three groups.

Routing

Band

The band that drives the effect, or **None** to hold it static.

Mode

Follow tracks the level; the other modes derive from the tempo.

INV

Inverts the response, so loud passages reduce the effect.

Response

Intensity	The effect's overall strength. Accepts keyframes.
Reactivity	How much of that strength the audio controls. At 0 % the effect holds static.
Attack / Release	Per-effect override of the engine timings.
Curve	Shapes the response, from linear to sharply exponential.
Bounce	Adds overshoot, so a hit springs past its target and settles.

Parameters

The effect's own controls, each with a keyframe diamond. Chapter 28 lists them for all 26 effects.

Setting the look with **Intensity** first, and then raising **Reactivity**, gives a predictable result.

Audio

The program plays a mix, places and trims it, and drives the reactive engine from it.

21.1 Tracks and clips

Audio tracks sit at the foot of the track column and carry a subtle tint in the header and the row. Audio clips trim, move and fade like any other clip, and the waveform is drawn at screen resolution for the visible span.

21.2 The audio inspector

Waveform

A miniature of the clip.

Volume

Clip gain.

Wave scale

Vertical zoom of the waveform display, from $\frac{1}{4}\times$ to $8\times$. This is a view preference for the whole timeline and leaves the sound and the export as they are.

21.3 Fades

Audio fades use the clip's corner handles, and the fade is the volume.

21.4 Linked audio and video

Video imported with sound arrives as a linked pair, which moves and trims together. The clip menu unlinks them.

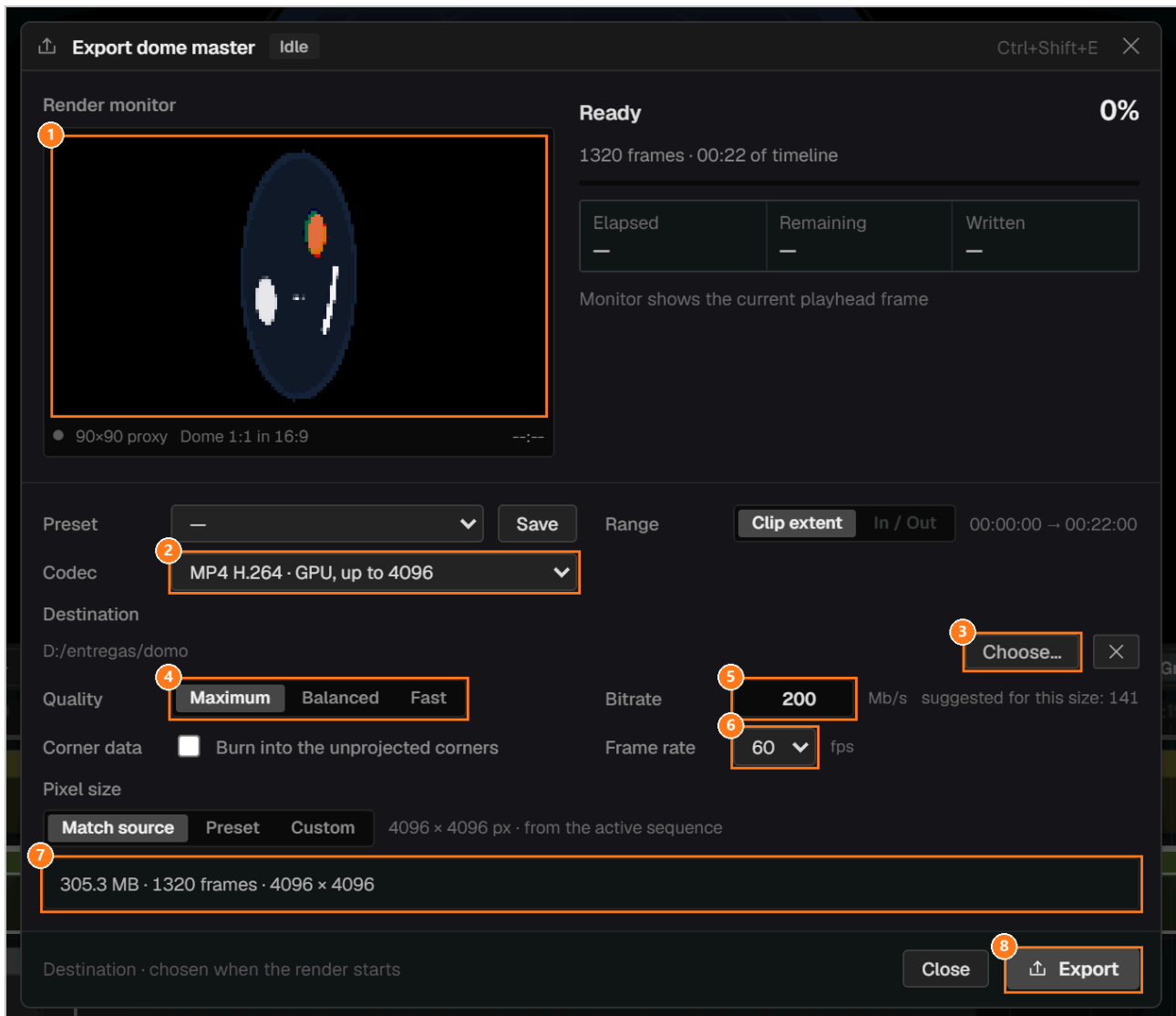
21.5 Audio in exports

- **MP4** carries stereo audio inside the file.
- **Other codecs** — PNG sequence, HAP — receive audio as a separate file.

21.6 Beat markers

A command in the palette detects beats in an audio clip and places locators on them, which then serve as snap targets.

Exporting



The export sheet. It is draggable, and the estimate follows each change.

- 1 **Render monitor** — the frame the encoder has just written.
- 2 **Codec** — the format and encoder, listed below.
- 3 **Destination** — Choose... sets the folder before the render starts.
- 4 **Quality** — Maximum, Balanced or Fast.
- 5 **Bitrate** — in Mb/s, with a suggested figure for the chosen size.
- 6 **Frame rate** — the output rate.
- 7 **Estimate** — file size, frame count and dimensions.
- 8 **Export** — starts the render.

Above these, **Range** selects **Clip extent** or the marked **In / Out** range, and **Pixel size** takes the dimensions from the sequence with **Match source**, from a preset, or from typed values.

22.1 Codecs

CODEC	MAXIMUM SIZE	USE
PNG sequence · lossless	—	Mastering and archival, with or without alpha. Without alpha the background is black and the result stays lossless.
MP4 H.264 · GPU	4096 × 4096	A 4K dome master.
MP4 H.265 · GPU	8192 × 8192, 10-bit	Large formats, wide room strips, and 10-bit material.
MP4 · H.264	3072 × 3072	The built-in encoder.
MP4 · H.265	1080 × 1080	The built-in encoder.
MOV · HAP / HAP Q	—	Playback systems that use HAP. Large files, very cheap to decode.
Still frame · PNG	—	A single frame.

The two H.264 entries use different encoders. The entries marked **GPU** use the graphics card's hardware encoder and reach dome resolutions. The other two use the built-in encoder, which reaches 3072 × 3072 for H.264 and about 1080 × 1080 for H.265.

22.2 Quality and bit depth

For a 4096 × 4096 master, 150 to 250 Mb/s at **Maximum** covers delivery. **Bit depth** offers 10-bit on H.265, which suits the large smooth gradients that appear across a dome.

22.3 Presets

Save beside the **Preset** menu stores a configuration for reuse — one per venue, for example.

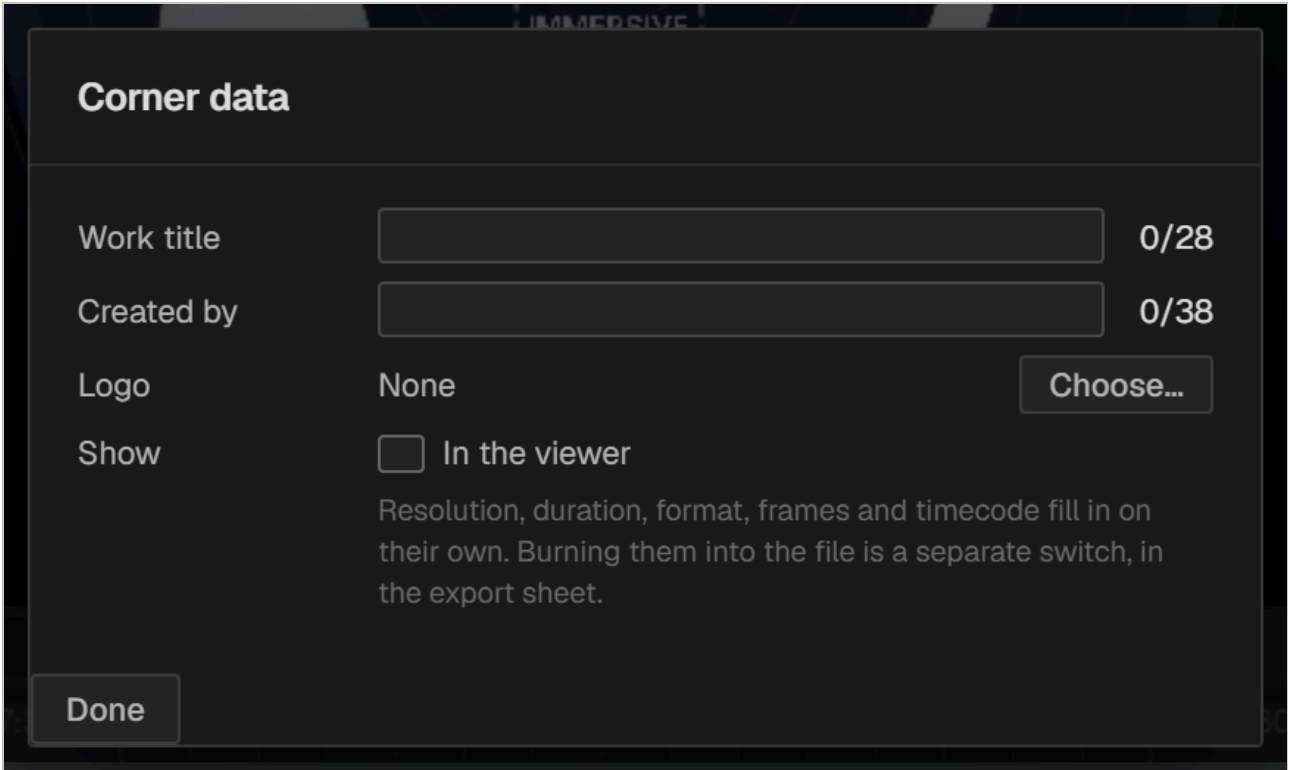
22.4 The queue

Jobs run one at a time. A room export in **Each wall + floor** mode queues one job per surface. Elapsed, remaining and written counters sit beside the render monitor.

A 4096 × 4096 export is a long operation. Running it for a minute, stopping, and opening the file in the player that will be used confirms the settings before the whole piece is committed.

Corner data

A dome master is a circle inside a square, which leaves four corners outside the projected area. Corner data places identification there: title, author, and a running timecode.



Corner data settings, from **Window** → **Corner data settings...**

23.1 Positions

Bottom left	The title of the work, in large type.
Bottom right	Timecode and frame count, advancing with the export.
Top left	A logo image.
Top right	Creator, resolution, duration and format.

23.2 Viewing and recording

Window → **Show corner data** draws it in the viewer, as a view setting.

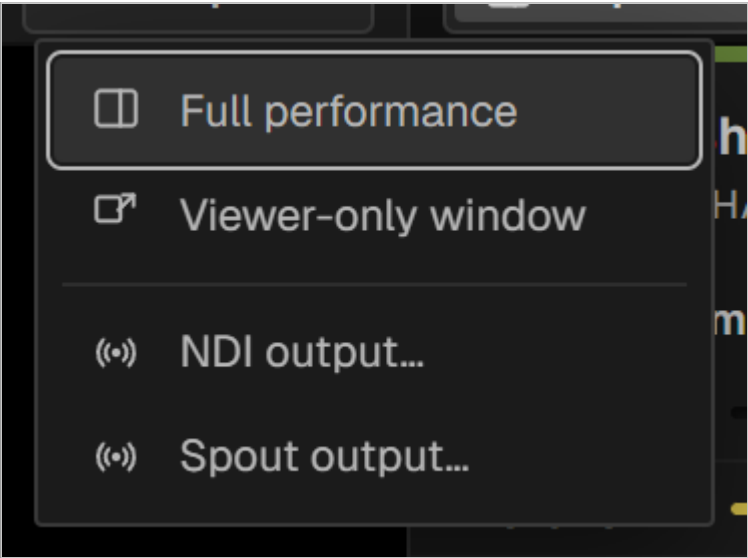
Corner data → **Burn into the unprojected corners** in the export sheet records it into the file. The two switches are independent: an exported master carries corner data when the export switch asks for it.

23.3 Details

- The timecode counts exported frames, so it matches the delivered file.
- Text blocks shrink automatically when a long title would reach the projected circle.
- In a dome sequence the picture is clipped to the circle before the corner data is drawn.
- An optional black background fills the corners for players that expect opaque frames.

Live output

The composite master can be broadcast while work continues, which drives a dome during a rehearsal or feeds a media server.



The Output menu, above the viewer. A pulsing dot marks an active broadcast.

Full performance	The viewer occupies the whole window.
Viewer window	A second window for another display, showing the complementary view.
NDI	Broadcasts the clean master over the network to any NDI receiver.
Spout	Shares the master with another application on the same Windows machine, GPU to GPU.

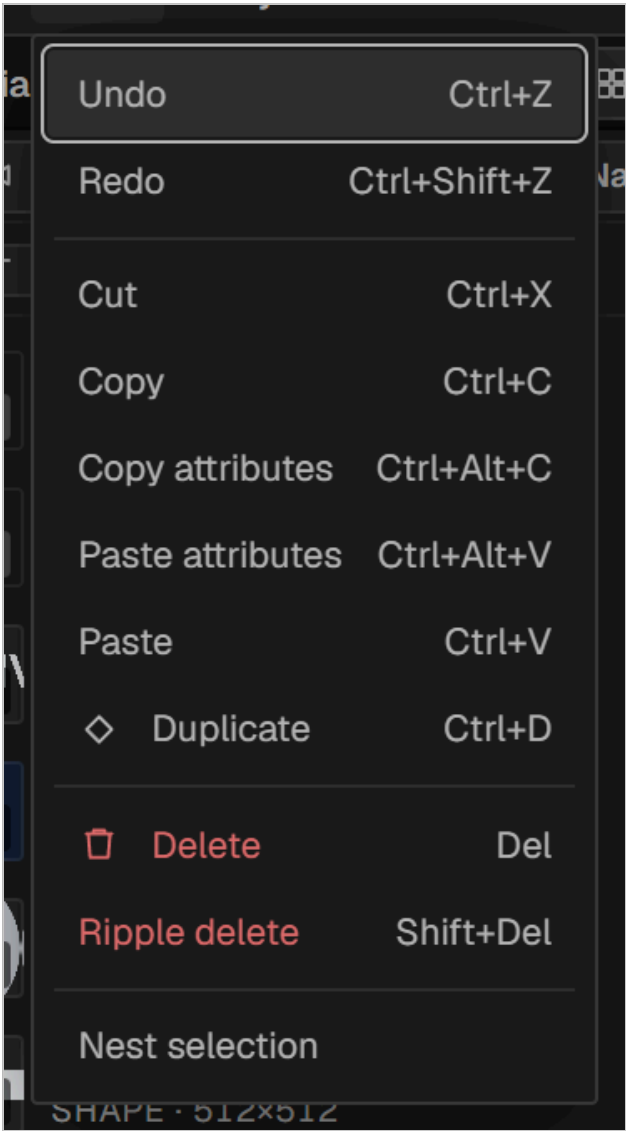
The broadcast carries the clean master: guides, overlays and corner data stay in the editor.

24.1 NDI input

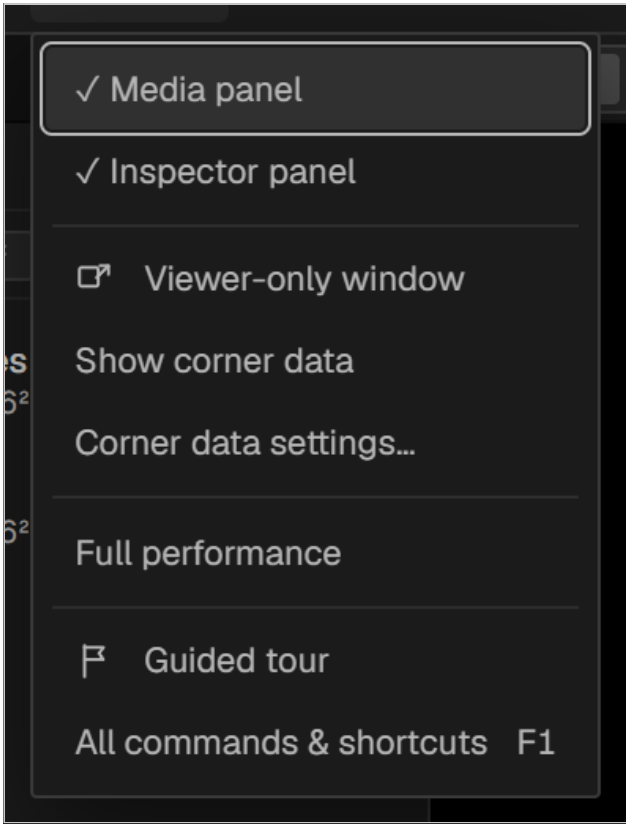
A live NDI stream can be added as a media item and used as a clip, which brings in a live camera or another machine's output.

NDI requires its runtime, and the program offers a link to it when it is absent. Spout runs on Windows.

Menus



Edit.



Window.

25.1 File

New project...	Ctrl + N	Opens the start screen; the current project stays open behind it.
Open...	Ctrl + O	Opens .isp, and .ise and .rdome from earlier versions.
Save	Ctrl + S	
Save As...	Ctrl + Shift + S	
Export...	Ctrl + Shift + E	

25.2 Edit

Undo / Redo	Ctrl + Z	
Cut	Ctrl + X	
Copy / Paste	Ctrl + C · Ctrl + V	Operates on the whole selection.
Copy attributes	Ctrl + Alt + C	Chapter 11.
Paste attributes	Ctrl + Alt + V	
Duplicate	Ctrl + D	
Delete	Del	Leaves a gap.
Ripple delete	Shift + Del	Closes the gap.
Nest selection		Chapter 15.

25.3 Project

Sequence settings...		Resolution, frame rate, dome coverage.
Dome coverage (FOV)...		Dome sequences.
Room geometry...		Room sequences. Reconfigures the active room.
New sequence...		Dome, 2D or 360 Room.
Preferences...	Ctrl + ,	

25.4 Window

Media panel		Shows or hides it.
Inspector panel		Shows or hides it.
Viewer-only window		Second window, complementary view.
Show corner data		Draws it in the viewer.
Corner data settings...		Title, author and logo.
Full performance		Viewer fills the window.
Guided tour		Restarts the tour.
All commands & shortcuts	F1	The command palette.

Keyboard shortcuts

On macOS, **Ctrl** means **⌘**. The complete list is available in the program with **F1**.

Transport

Space	Play / Pause
Home	Go to start
End	Go to end
J	Shuttle back (J · press again = 2×/4×/8×)
L	Shuttle forward (L · press again = 2×/4×/8×)
K	Stop shuttle

File

Ctrl + N	New project
Ctrl + S	Save
Ctrl + I	Import media...
Ctrl + Shift + E	Export master...
Ctrl + O	Open project
Ctrl + ,	Preferences...
Ctrl + Shift + S	Save As... (new file)

Edit

Ctrl + Z	Undo
Ctrl + Shift + Z	Redo
Ctrl + D	Duplicate
Del	Delete
Shift + Del	Ripple delete
Ctrl + C	Copy
Ctrl + V	Paste

Timeline

Ctrl + E	Split at selection / playhead
Alt + ←/→	Nudge clip ±1 frame / ±1 s
+	Zoom in
-	Zoom out
↑	Previous cut
↓	Next cut

Locators

M	Add locator
.	Next
,	Previous

Tools

V	Select
H	Hand / Pan
B	Razor
Z	Zoom

Compose parameters

27.1 Dome types

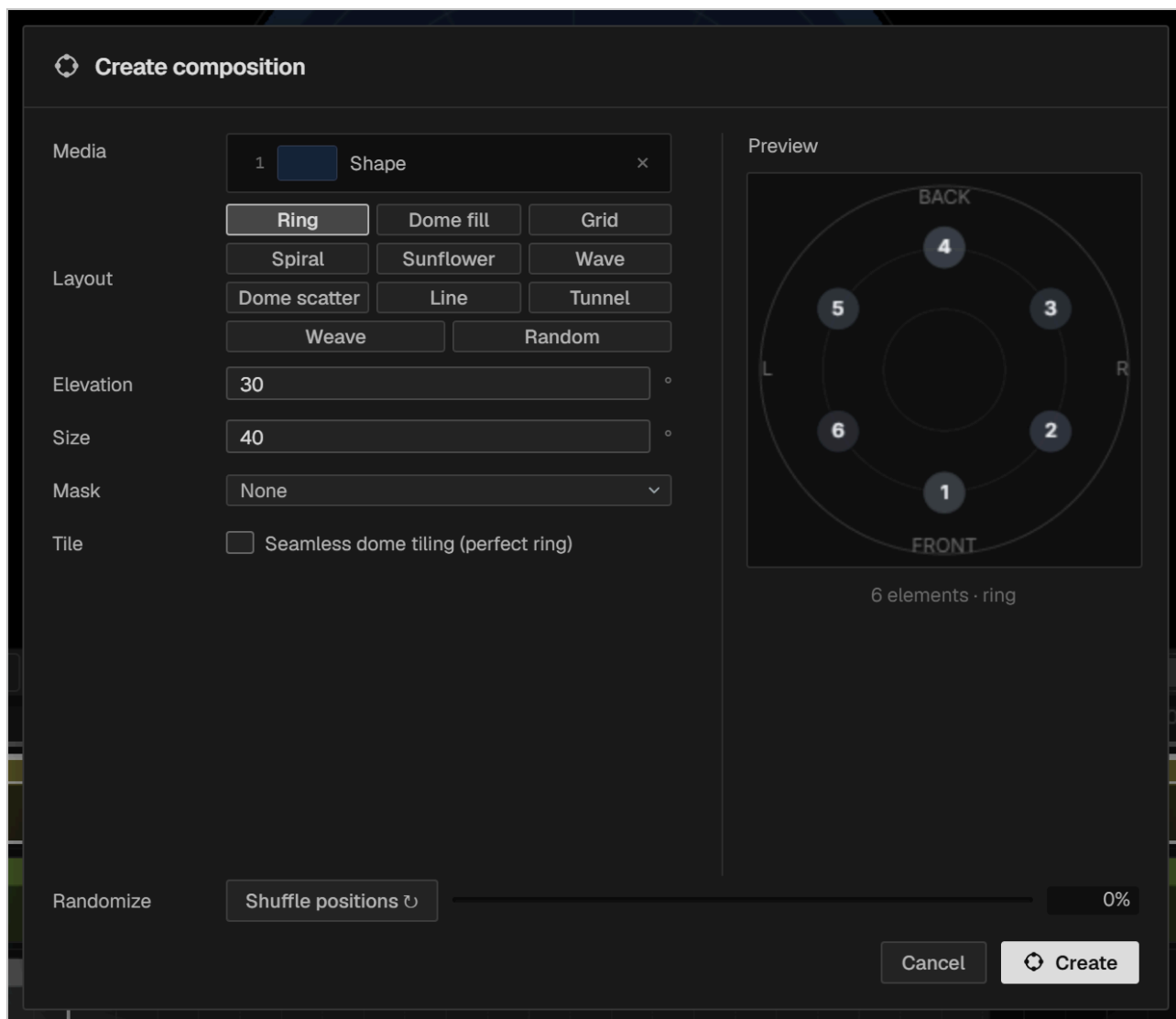
TYPE	PARAMETERS
Ring	Count, elevation, size, ring radius.
Dome fill	Rings, segments, ring gap, segment gap, offset, tiles.
Spiral	Count, turns, size.
Sunflower	Count, size.
Wave	Count, amplitude, frequency.
Dome scatter	Count, size, jitter.
Random	Count, size, jitter.
Tunnel	Count, from, to, speed, depth, spin, helix, fade in, fade out.

27.2 Flat types

TYPE	PARAMETERS
Grid	Columns, rows, gap.
Line	Count, spacing, angle.
Row / Column	Count, spacing.
Weave	Layout, strips, strip width, packing, long side, direction, speed per family, order.

27.3 Common parameters

Media	The basket: which sources, and in what order.
Mask	Shape and size, 0 % to 100 %, applied to every element.
Order	Shuffles which source goes where.



Ring.

Effects

Twenty-six effects, grouped by family. Every parameter accepts keyframes, and every effect accepts audio drive from the Reactive FX tab. Effects marked **feedback** read the previous frame, so their result builds over time.

Distortion

EFFECT	PARAMETERS
Scale Pulse	Amount
Rotate	Angle · Spin
Glitch	Blocks · RGB · Rate
Datamosh · feedback	Smear · Dir X · Dir Y
Slice	Slices · Rate
Pixelate	Blocks
Kaleidoscope	Segments · Angle · Zoom
Mirror	Mode 0-4
Wave	Amount · Frequency · Speed
Zoom Blur	Strength · Center X · Center Y
Noise Warp	Scale · Amount · Speed

Stylize

EFFECT	PARAMETERS
Strobe	Flash · Hardness
Edge	Thickness · Mix
Posterize	Levels
Scanlines / CRT	Lines · Strength · Chroma
Bloom / Glow	Threshold · Radius · Boost

Colour

EFFECT	PARAMETERS
--------	------------

RGB Split	Angle · Spread
Hue Shift	Shift
Chroma Pulse	Strength · Center X · Center Y
Flash	Mode W/B/Inv

Feedback

EFFECT	PARAMETERS
Trails / Echo · feedback	Decay
Feedback Zoom · feedback	Zoom · Rotate · Decay
Feedback Flow · feedback	Zoom · Rotate · Hue Cycle · Warp · Decay

Dome

EFFECT	PARAMETERS
Dome Rings	Rings · Sharpness · Travel
Spiral Twist	Twist · Rotation
Tunnel	Depth · Flight

Performance and problem solving

29.1 Smooth playback

1. **Generate proxies for heavy footage.** With long-GOP camera masters this gives the largest gain.
2. **Lower the viewer quality** to $\frac{1}{2}$ or $\frac{1}{4}$, which leaves the export untouched.
3. **Cache heavy compositions** with a composition proxy, so playback decodes one stream.
4. **Render in place** anything finished and expensive.
5. **Mute tracks** that are not being judged.

29.2 Situations

An element inside a composition looks softer than the same footage placed directly

Compositing passes through an intermediate buffer whose resolution follows the output size. Exporting at the sequence's own resolution keeps the detail.

Part of a clip appears in the corners of the 2D view

That area falls outside the fulldome circle and records as black, or as transparency with alpha. Reducing the clip's size or moving it brings it inside.

An export refuses a resolution

The built-in H.264 reaches 3072×3072 and the built-in H.265 about 1080×1080 . The entries marked GPU reach 4096 and 8192.

A clip plays every second frame

Its proxy was generated at half the source's frame rate. Deleting the proxy and generating it again restores the rate.

A clip shortened when looping was switched off

Switching looping off trims the clip to the remaining length of the file. **Ctrl** + **Z** restores it, and **From clip** redefines a cycle without switching the loop off.

A project opened with an unexpected format or framing

Factory defaults apply before every project is read. **Project** → **Sequence settings...** shows the sequence's current values.

The 3D view is black

On a hybrid-graphics laptop, the program requests the discrete GPU automatically; a system-level override can redirect it.

29.3 Recovering work

- Projects save atomically with a `.bak` alongside.
- `Restore last autosave` and `Recovery history...` are in the command palette.
- `Save incremental` writes `_v01`, `_v02` and so on, which suits the moment before a large restructuring.

29.4 Reporting a problem

`Save diagnostics log...` in the command palette writes a log describing the environment and recent activity.